

Government Reputation, FDI, and Profit Shifting*

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Abstract

I document that higher investment risk is associated with higher statutory corporate tax rates within countries. I study this relationship in a model in which the government announces a tax before firms invest but may raise it afterward. Firms can protect part of their returns through profit shifting. In the static model, I show that this protection makes the optimal announced tax fall as government reputation improves. Without profit shifting, a better reputation instead leads to a higher tax. In a dynamic extension of the model, profit shifting induces an optimal tax announcement decrease in government reputation, as seen in the data, whereas the optimal tax is nonmonotonic in an economy with no profit shifting. Profit shifting also helps explain why investment and production persist in economies with weak policy credibility, sustaining both even as reputation approaches zero, whereas investment vanishes when there is no profit shifting.

Keywords: Commitment, Corporate taxation, Foreign direct investment, Government reputation, Multinational firm, Profit shifting

JEL codes: H21, H25, H26, F21, F23

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1 Introduction

A government has an incentive to raise corporate taxes after firms have invested. Firms anticipate this incentive and invest less, reducing the capital available for production and taxation. A reputation for implementing announced policies can mitigate this problem, but firms can also protect their returns by reporting profits abroad. Profit shifting therefore affects the return to investment even when the government cannot commit to its tax policy. This paper studies how this protection changes the relationship between government reputation, corporate taxation, and investment.

I begin by documenting that higher investment risk is associated with higher statutory corporate tax rates within countries. I use OECD tax data and measure risk with the ICRG Investment Profile, which covers the security of contracts, protection from expropriation, profit repatriation, and payment delays. The association remains positive and significant after controlling for economic conditions and other institutional and political risks. With these controls, a one-point increase in risk is associated with a 0.43 percentage-point higher tax rate. I also examine foreign affiliates' investment around the 2018 *Achmea* judgment, which restricted investor–state arbitration under intra-EU treaties. The estimates indicate lower investment in covered pairs following the judgment. These findings motivate the analysis of taxation and investment when firms are uncertain about the protection of their returns.

I then develop a small open economy model in which multinational firms allocate capital between a host country and a foreign location. The host government announces a statutory tax rate before firms invest and implements its tax afterward. Firms then choose where to report profits. The government is either a commitment type, which must implement its tax announcement, or an opportunistic type, which can revise the tax after investment. Reputation is the probability firms assign to the commitment type. Following [Lu \(2013\)](#), I study common tax announcements and select the commitment type's preferred rate subject to both types' incentive compatibility constraints.

In the static model, profit shifting limits the revenue the government can collect from installed capital. Firms respond to a higher implemented tax by reporting fewer profits in the host country, preserving part of their returns under discretion. They therefore invest even when they know the government cannot commit. Better reputation raises investment at a given announced tax rate and puts greater weight on the announced rate in firms' investment decisions. The government can tax a larger capital stock, but a higher announced tax also discourages investment. I derive a condition under which the return firms retain under discretion makes the proportional investment loss from a tax increase rise with reputation. The commitment type then chooses a lower tax. For a class of strictly convex profit shifting costs deducted in the country from which

profits are transferred, the optimal tax is unique and decreases with reputation, while investment and tax revenue increase.

Without profit shifting, the opportunistic type taxes all host profits. Firms receive a positive return to host investment only when the government can commit. Better reputation then expands investment sufficiently for the commitment type to announce a higher tax. Profit shifting reverses this tax response under the stated conditions, holding the production technology fixed. Numerical comparisons also give a declining announced tax when profit shifting costs are incurred in both countries or are not deductible. Thus the return firms retain after taxation provides a mechanism through which a more credible government chooses a lower statutory rate, consistent with the empirical relationship between investment risk and taxation.

In a dynamic extension of the model, a tax revision reveals that the government cannot commit and reduces future investment and tax revenue. The opportunistic type weighs this future loss against the current revenue gain from a tax increase. Firms anticipate its implementation decision, and the commitment type chooses its tax announcement taking both current investment and future revenue into account. With profit shifting, I establish that the optimal announced tax decreases with reputation when reputation is sufficiently low. The return firms retain under discretion preserves a positive investment base even when commitment is unlikely. The declining tax response therefore extends to a setting in which the opportunistic type can choose to implement its tax announcement.

I then compare optimal announced taxes and real activity numerically with and without profit shifting. Without profit shifting, the tax first rises and then falls as reputation improves. The increase occurs while tax implementation remains uncertain. The decline begins once both types implement the preferred announced tax rate. With profit shifting, the tax declines until the full-commitment rate is attained. Profit shifting therefore preserves the direction of the static tax response in this comparison. It also reduces the investment lost under uncertain commitment. At a reputation of one half, investment is 52 percent below its full-commitment level without profit shifting and 4 percent below with profit shifting. The comparison holds the production technology and discount factor fixed. Protecting firms' returns thus matters for both the tax the government announces and the real activity that takes place under that policy.

I next characterize sustainable taxes, which both government types implement. At a given sustainable tax, current investment is independent of reputation because firms face the same rate under either type. Better reputation instead raises the future revenue lost after a tax revision, allowing a lower tax to be sustained. This mechanism operates with or without profit shifting. It characterizes the optimal announced tax rate when the government chooses a tax that both types implement. The choice among all tax

announcements also accounts for the commitment type's gain in future reputation when the opportunistic type sometimes revises the tax. This additional consideration explains why the optimal announced tax and the optimal sustainable tax can differ.

Finally, I extend the analysis of sustainable taxation and investment to longer finite horizons. The incentive to implement a tax now depends on the revenue lost in all remaining periods. Over any fixed finite horizon, investment without profit shifting converges to zero as reputation approaches zero, and host production vanishes with it. With profit shifting, investment and production remain positive because firms retain part of their returns after taxation. This difference in real activity also holds when tax implementation and future tax announcements respond to the observed policy history. Profit shifting therefore allows the model to account for investment and production in economies with weak policy credibility. The return that changes the optimal tax in the static model also sustains real activity when reputation alone provides little protection.

The closest comparison for the static result is [Mardan and Stimmelmayer \(2020\)](#), who show that profit shifting can reverse the relationship between country risk and corporate tax rates. Their risk is an exogenous probability of project failure that destroys the investment return. They distinguish it from expropriation that generates government revenue. Here the government chooses the tax revision whose risk firms anticipate. Installed capital continues to produce after this choice, while firms adjust reported profits. I characterize how the return retained under discretion connects this reporting response to the optimal tax announcement. The same investment decision determines the future revenue at stake when the government can preserve its reputation by implementing the tax announcement.

The analysis also builds on the distinction between real investment and reported profits. Profit shifting reduces the effective tax burden on mobile capital in [Hong and Smart \(2010\)](#) and affects tax competition in [Peralta et al. \(2006\)](#), [Devereux et al. \(2008\)](#), [Johannesen \(2010\)](#), and [Mongrain et al. \(2023\)](#). Firms can protect returns after investment through excess foreign capacity ([Janeba, 2000](#)), private information about intrafirm trading costs ([Konrad and Lommerud, 2001](#)), and mobility ([Langenmayr and Simmler, 2021](#)). I study how the protection from profit shifting changes the tax announced before investment when the government's ability to commit is uncertain. I then show how the same protection affects investment when future revenue disciplines tax implementation.

The dynamic analysis relates to capital taxation under limited commitment ([Chari et al., 1988](#); [Thomas and Worrall, 1994](#); [Aguiar and Amador, 2011](#); [Schnitzer, 1999](#)). [Chari and Kehoe \(1990\)](#) characterize sustainable plans by sequential government and private-sector incentives. Their capital-tax example distinguishes the zero-investment finite-horizon outcome from the larger set of sustainable infinite-horizon outcomes. I study

how uncertainty about commitment and the mobility of reported profits affect taxation and investment over a finite horizon. Without profit shifting, investment vanishes as reputation approaches zero even when the opportunistic type implements its tax announcement with positive probability. Profit shifting preserves investment through the return firms obtain after taxation. This protection does not require a particular future punishment and extends beyond policies implemented by both government types.

The tax announcement problem follows [Lu \(2013\)](#). It differs from the choice of a rule by a designer uncertain about the policy maker's commitment type in [Dovis and Kirpalani \(2021\)](#). Under their concavity condition, uncertainty about commitment is beneficial and a rule can optimally induce implementation by both types. Their foreign-investment example shows that the required concavity in reputation need not hold. Here the commitment type chooses its own tax announcement and can prefer a tax that the opportunistic type sometimes revises. I characterize the tax response at low reputation and the finite-horizon investment result allowing for this choice. These results connect firms' reporting decisions to tax policy and real investment when implementation is endogenous.¹

The paper is organized as follows. Section 2 presents the empirical evidence. Section 3 introduces the model, and Section 4 studies optimal taxation with and without profit shifting. Section 5 first examines optimal tax announcements and investment when tax implementation is endogenous, then studies sustainable taxation and investment over longer horizons. Section 6 concludes.

2 Investment Risk and Corporate Taxation

I examine the relationship between investment risk and statutory corporate tax rates within countries, controlling for economic conditions and other institutional and political risks. I then study foreign affiliates' investment following a change in treaty protection.

2.1 Investment Risk and Tax-Policy Credibility

The credibility relevant to the model concerns the return on capital after it has been installed. Firms' investment decisions depend on whether they expect the government to implement its announced statutory rate. Once capital is fixed, an additional tax can transfer part of its return to the government. Constraints on the government's ability to

¹International agreements offer another source of investment protection ([Kohler and Stähler, 2019](#); [Horn and Tangerås, 2021](#)).

appropriate private returns therefore affect the credibility of its tax policy (North and Weingast, 1989; Acemoglu and Johnson, 2005).

I use the ICRG Investment Profile to measure risks to those returns. It combines Contract Viability/Expropriation, Profits Repatriation, and Payment Delays, with four points assigned to each component and higher scores denoting lower risk (The PRS Group, nd). The contract and expropriation component is most closely related to the model. It concerns the security of returns under arrangements made before investment, when firms can still choose where to locate capital. A tax increase after investment can appropriate part of the same return. Firms account for this possibility before investing, which connects the security of private returns to tax-policy credibility.

The other components concern firms' ability to receive and transfer their returns. They make Investment Profile relevant to the treatment of foreign investment, although they are broader than taxation. This interpretation has precedent in World Bank (2004, pp. 49–50, 178), which uses Investment Profile when discussing constraints on discretionary policy and the credibility of commitments to firms. The report also connects contractual commitments to tax policy and international arbitration. These links support using the index as a proxy for the security of returns against subsequent government intervention.

I define investment risk as

$$\text{Risk}_{ht} = 12 - \text{Investment Profile}_{ht},$$

so that higher values indicate greater risk. The score is averaged across months and lagged one year in the regressions. Because it measures the security of investment returns more broadly than tax-policy credibility, I do not equate it with the model's probability of commitment.

2.2 Statutory Tax Rates

I use the OECD combined statutory corporate income tax rate, including central and subnational taxation (OECD, 2026). The model counterpart is τ^R , the statutory rate announced before investment. A subsequent tax revision changes this rate to τ^O . The observed annual rate does not distinguish tax announcements from subsequent tax revisions.

TABLE 1: Investment risk and statutory tax rates within countries

	(1)	(2)	(3)	(4)	(5)
Investment risk	0.395	0.394	0.435	0.426	0.420
Standard error	(0.148)	(0.149)	(0.142)	(0.143)	(0.145)
<i>p</i> -value	0.009	0.009	0.003	0.004	0.005
Economic controls	Yes	Yes	Yes	Yes	Yes
Institutional controls	No	Yes	No	Yes	Yes
Political controls	No	No	Yes	Yes	Yes
Lagged GDP growth	No	No	No	No	Yes
Country and year effects	Yes	Yes	Yes	Yes	Yes
Observations	1,887	1,887	1,887	1,887	1,887
Countries	101	101	101	101	101
Within R^2	0.025	0.031	0.033	0.038	0.039

Notes: OECD annual rates over 2003–2021. All columns use the same 1,887 observations in 101 countries and include country and year effects. Economic controls are log real GDP per capita, log population, and resource rents/GDP. Institutional controls are corruption, law and order, and bureaucratic quality. Political controls are government stability, internal conflict, and external conflict. All ICRG controls are lagged risks. Parentheses contain country-clustered standard errors with 100 reference degrees of freedom. Reported R^2 is within R^2 . Appendix Table 5 gives all slope coefficients for columns (4) and (5).

The sample contains 1,887 country-year observations in 101 countries over 2003–2021.² I estimate

$$\text{Tax rate}_{ht} = \alpha_h + \lambda_t + \beta \text{Risk}_{h,t-1} + \delta' X_{ht} + \phi' Z_{h,t-1} + u_{ht}. \quad (1)$$

The economic controls in X are log real GDP per capita, log population, and natural-resource rents as a share of GDP. They account for development, market size, and resource dependence. Country effects absorb persistent differences across countries, and year effects absorb common changes in tax policy. The vector Z contains other ICRG measures. One block includes corruption, law and order, and bureaucratic quality. Another includes government stability, internal conflict, and external conflict. The first accounts for institutional conditions that may affect both risk assessments and tax collection. The second accounts for political disruptions that may affect investment and fiscal policy independently of the treatment of installed capital. Investment Profile is a separate component in the ICRG methodology (The PRS Group, nd). Including these measures helps distinguish its association with taxes from broader institutional and political conditions. All ICRG variables are expressed as risks, with higher values indicating worse conditions, and are lagged one year.

The coefficient on investment risk is 0.395 with the economic controls ($p = 0.009$). It remains positive and significant when institutional controls, political controls, or both are added on the same sample. With all controls, a one-point increase in risk is

²I exclude twenty zero-rate observations inconsistent with historical tax descriptions. Appendix C.1 documents the exclusions and reports estimates retaining the original records.

associated with a statutory rate higher by 0.426 percentage points ($p = 0.004$). Adding lagged real GDP growth gives 0.420 ($p = 0.005$). The association therefore survives these controls for development, institutions, political disruptions, and the business cycle.

The positive association also holds under alternative treatments of the source records (Appendix Table 6). These estimates describe conditional associations rather than the causal effect of tax-policy credibility.

Independent firm-level evidence informs the timing of the model. [Hopland et al. \(2018\)](#) document adjustments in intragroup payments within the tax year in Norwegian tax records. [de Mooij and Liu \(2020\)](#) find a relative decline in multinational affiliates' investment following transfer-pricing rules. The model allows reported profits to respond after real investment is installed, while anticipated reporting opportunities affect the return to investment.

2.3 Investment Following a Change in Treaty Protection

The decision in *Achmea* changed the enforcement of investment protection under intra-EU treaties. On March 6, 2018, the Court of Justice of the European Union ruled that the investor–state arbitration clause in an intra-EU bilateral investment treaty was incompatible with EU law ([Court of Justice of the European Union, 2018](#)). Firms covered by such treaties consequently faced a change in how they could enforce protection against their host governments. I examine their investments around the judgment.

The investment data come from Eurostat's inward foreign affiliates statistics ([Eurostat, 2026](#)). They report gross investment in tangible goods by foreign-controlled enterprises in millions of euros for each host and ultimate-parent country pair over 2010–2019. I identify treaty coverage just before *Achmea* from the lists attached to the termination agreement ([European Union, 2020](#)). For inclusion, both the host and the parent country must be parties to that agreement. I drop pairs whose treaties had already been terminated. There are 22 host and 22 parent countries in the resulting panel.

I estimate annual investment coefficients by Poisson pseudo-maximum likelihood, with country-pair, host-year, and parent-year fixed effects. These absorb persistent differences across pairs and annual conditions in each host and parent country. A second specification includes a separate linear time trend for each host-parent country pair. Standard errors are clustered by host and parent. Appendix C.3 reports the equation and annual estimates.

The mean of the 2018 and 2019 coefficients is -0.474 without country-pair linear trends and -0.648 with them (Table 2). The estimates suggest lower investment in

TABLE 2: Investment after the Achmea judgment

	(1)	(2)
Average post-judgment coefficient	−0.474** (0.215)	−0.648** (0.280)
<i>p</i> -value	0.039	0.031
Country-pair linear trends	No	Yes
Pre-period joint <i>p</i> -value	0.017	0.549
Observations	2,622	2,547
Host and parent countries	22 each	22 each
Country pairs	349	320

Notes: Each column reports an annual event-study regression estimated by Poisson pseudo-maximum likelihood over 2010–2019. The post-judgment coefficient averages the 2018 and 2019 coefficients. Both regressions include pair, host-year, and parent-year effects and use 2017 as the reference year. Column (2) also includes a separate linear time trend for each country pair. Parentheses contain standard errors clustered by host and parent country, using 21 reference degrees of freedom. The joint pre-period tests use host-clustered covariance matrices. ***, **, and * indicate significance at 1, 5, and 10 percent.

covered pairs after the judgment. Their magnitude and precision depend on reporting coverage and pre-event investment trends, so they do not establish a causal effect of reduced treaty protection. Appendix C.3 reports the annual estimates and checks using alternative coverage and trend specifications.

Host-year effects absorb annual changes in host tax policy, so the comparison concerns treaty protection rather than an observed tax revision. The model studies how the risk of a tax revision affects firms’ investment and reporting choices and the statutory tax chosen before investment.

3 Model

The government announces a corporate tax rate before firms invest and sets the implemented rate after capital is installed. Firms then choose where to report their profits. The commitment type must implement its tax announcement, whereas the opportunistic type can reoptimize, taking firms’ profit-shifting responses into account. I solve the model by backward induction.

3.1 Environment and Timing

There is a unit measure of identical, foreign-owned multinational firms. The host government applies one common tax rate, with no firm-specific taxation. Each firm allocates a fixed capital endowment $\bar{k} > 0$ between the host country and a foreign location. Investment k in the host produces profits $\rho(k)$, while investment $\bar{k} - k$ abroad produces $\rho(\bar{k} - k)$. Profits are net of the rental cost of capital. The profit function

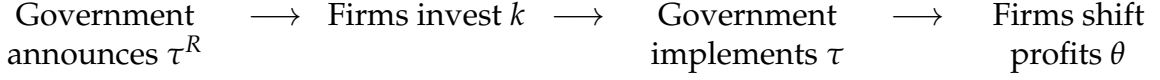


FIGURE 1: Timing within a period

satisfies $\rho(0) = 0$ and is increasing and strictly concave on the investment domain. The parameterized comparisons use $\rho(k) = zk^\alpha - r^*k$, where $z > 0$, $0 < \alpha < 1$, and $r^* \geq 0$ is the rental rate. In this specification, zk^α is output produced in the host and r^*k is the cost of capital. The host is a small open economy that takes the foreign corporate tax rate $\tau^* \in (0, 1)$ as given.

There are two government types, both of which maximize host corporate tax revenue. The commitment type must implement the tax announced before investment. The opportunistic type can reoptimize after firms invest, taking installed capital as given. Firms initially do not observe the government's type and assign probability p to the commitment type. I refer to this probability as government reputation.

Write τ^R for the statutory tax rate announced before investment, which the commitment type implements. The realized host rate is $\tau \in [0, 1]$. Firms choose their profit-shifting amount θ after observing τ . A positive θ moves profits into the host country and a negative θ moves profits out. All tax announcements and implemented rates are public. The choices occur in the following order.

Capital is fixed for the remainder of the period once it has been invested. Intragroup payments can still be changed within the firm's existing structure before taxes are filed, consistent with the evidence in Section 2.2. I begin with that reporting decision.

3.2 Stage 4. Firms' Profit-Shifting Choice

Given its capital and both tax rates, the firm chooses θ to maximize its total after-tax profit,

$$\begin{aligned} \max_{\theta} \quad & (1 - \tau) \left[\rho(k) + \theta - \frac{\gamma\theta^2}{2\rho(k)} \right] \\ & + (1 - \tau^*) \left[\rho(\bar{k} - k) - \theta - \frac{\gamma\theta^2}{2\rho(\bar{k} - k)} \right], \\ \text{subject to} \quad & \rho(k) + \theta - \frac{\gamma\theta^2}{2\rho(k)} \geq 0, \\ & \rho(\bar{k} - k) - \theta - \frac{\gamma\theta^2}{2\rho(\bar{k} - k)} \geq 0. \end{aligned} \tag{2}$$

The resources used in arranging transfers, such as legal and accounting services, are governed by $\gamma > 0$. Costs are deductible in each country. I require both reported tax bases to be nonnegative, including when the host taxes profits at a rate of one.

Let $\theta(k, \tau)$ denote the firm's optimal transfer. When neither constraint binds,

$$\theta(k, \tau) = \frac{(\tau^* - \tau)\rho(k)\rho(\bar{k} - k)}{\gamma[(1 - \tau)\rho(\bar{k} - k) + (1 - \tau^*)\rho(k)]}. \quad (3)$$

The sign of (3) is the sign of $\tau^* - \tau$, so a host tax above the foreign rate induces an outward transfer. When this interior solution is outside the interval allowed by the two tax-base constraints, the firm's concave objective is maximized at the nearest endpoint of that interval. The bounds, derived in Appendix A.1, both tend to zero as investment in either location vanishes.

3.3 Stage 3. Optimal Taxation under Discretion

At this stage, aggregate host capital K is already installed. Write $\Lambda(K)$ for aggregate profits before profit shifting and $\Theta(K, \tau)$ for aggregate profit shifting. The unit measure of identical firms gives $\Lambda(K) = \rho(K)$ and $\Theta(K, \tau) = \theta(K, \tau)$ in a symmetric allocation. The commitment type implements τ^R . The opportunistic type chooses the tax that maximizes current revenue,

$$\tau^O(K) \in \arg \max_{\tau \in [0,1]} \left[\Lambda(K) + \Theta(K, \tau) - \frac{\gamma \Theta(K, \tau)^2}{2\Lambda(K)} \right]. \quad (4)$$

The opportunistic type trades off the tax rate against the reported base, taking installed capital as given. Unlike a government taxing immobile profits, it may collect more at a rate below one because firms move profits abroad in response to higher taxes.

3.4 Stage 2. Firms' Investment Choice

Firms invest after observing the tax announcement and before learning the implemented tax. When both government types announce the same rate, the tax announcement conveys no information about type. Firms then expect τ^R with probability p and $\tau^O(K)$ with probability $1 - p$. Each firm chooses capital to maximize

$$\begin{aligned} \max_{k \in [0, \bar{k}]} \mathbb{E}_\tau \left\{ (1 - \tau) \left[\rho(k) + \theta(k, \tau) - \frac{\gamma \theta(k, \tau)^2}{2\rho(k)} \right] \right. \\ \left. + (1 - \tau^*) \left[\rho(\bar{k} - k) - \theta(k, \tau) - \frac{\gamma \theta(k, \tau)^2}{2\rho(\bar{k} - k)} \right] \right\}. \end{aligned} \quad (5)$$

The expectation is over the tax rates imposed by the two government types. An atomistic firm takes them as given and does not internalize the effect of aggregate capital on the government's tax choice. Symmetry requires its chosen k to equal K . This

condition and (4) jointly characterize investment and discretionary taxation. I write $K(\tau^R, p)$ for capital in the equilibrium under consideration.

Profit shifting affects investment because it changes the return firms retain after taxation. Better reputation puts more weight on the rate under commitment in this comparison. A tax reduction can consequently attract more capital when firms consider it more likely to be implemented.

3.5 Stage 1. Optimal Taxation under Commitment

Following Lu (2013), I study the optimal tax announcement of a government that can commit but whose ability to do so is uncertain to firms. The tax announcement binds the commitment type, while the opportunistic type can imitate it and revise the tax after investment. I therefore consider common tax announcements and choose the one preferred by the commitment type, subject to both types' incentive compatibility constraints. Following an off-path tax announcement, firms believe the government is the opportunistic type with certainty. They then choose the investment appropriate for a government known to be the opportunistic type. Neither type can obtain more from such a tax announcement than the maximum tax revenue on that investment. A common tax announcement must give each type at least this amount.

For a given announced tax rate, tax revenue at reputation p and implemented rate τ is

$$R(p, \tau) = \tau \left[\Lambda(K) + \Theta(K, \tau) - \frac{\gamma \Theta(K, \tau)^2}{2\Lambda(K)} \right], \quad K = K(\tau^R, p).$$

In this notation, τ^R determines the investment on which revenue is collected. At $p = 0$, firms expect the discretionary tax regardless of the tax announcement. I use the same investment equilibrium with a government known to be the opportunistic type after every off-path tax announcement. Thus $\max_{\tau \in [0,1]} R(0, \tau)$ is the largest revenue either type can obtain after an off-path tax announcement.

The commitment type's preferred announced tax rate among these pooling equilibria solves

$$\begin{aligned} & \max_{\tau^R \in [0,1]} \tau^R \left[\Lambda(K) + \Theta(K, \tau^R) - \frac{\gamma \Theta(K, \tau^R)^2}{2\Lambda(K)} \right] \quad K = K(\tau^R, p), \\ & \text{subject to } R(p, \tau^R) \geq \max_{\tau \in [0,1]} R(0, \tau), \\ & R(p, \tau^O(K(\tau^R, p))) \geq \max_{\tau \in [0,1]} R(0, \tau). \end{aligned} \tag{6}$$

The two inequalities in (6) are the incentive compatibility constraints at the tax announcement stage. The first applies to the commitment type and the second to the

opportunistic type. The right-hand side is the payoff available following a tax announcement that induces belief zero. On the left-hand side, each type receives its equilibrium revenue at reputation p . For the opportunistic type this revenue is evaluated at its subsequent best response, $\tau^O(K(\tau^R, p))$, rather than at τ^R . Its implementation decision remains governed by (4).

The solution defines the tax under commitment at reputation p . Following Lu's approach to policy choice, I compare the commitment type's revenue across common tax announcements, keeping firms' belief equal to p . A deviation from a given equilibrium is evaluated at the stated off-path belief. Under the analytical assumptions below, any separating tax announcement outcome gives both types less revenue than the selected common tax announcement. Appendix A.2 verifies the incentive compatibility constraints, and Appendix A.7 gives the payoff comparison and its connection to Lu's equilibrium criterion.

When $p < 1$, commitment is private information and firms allow for a discretionary tax revision. The optimal tax under full commitment is the special case $p = 1$, when commitment is common knowledge. The government maximizes corporate tax revenue. A national welfare analysis would additionally require domestic factor incomes, ownership, and public expenditure.

4 Optimal Taxation under Commitment and Discretion

This section studies how reputation affects the announced tax rate, holding production, the foreign tax rate, and profit-shifting opportunities fixed. I first derive the tax response with profit shifting, then compare it with the benchmark without profit shifting. The difference depends on the after-tax profits firms receive under discretion. I characterize this mechanism beyond quadratic profit shifting costs and illustrate it numerically when costs are deductible in both countries. Section 5 turns to the government's incentive to implement its tax announcement.

4.1 Optimal Taxation with Profit Shifting

An increase in p changes both the level of investment and its sensitivity to τ^R . At a fixed $\tau^R < \tau^O$, the level rises because the expected after-tax payoff is higher. The commitment type can then tax a larger capital stock. The sensitivity changes because firms weight the payoff under τ^R by p in their investment condition. When choosing the tax announcement, the government trades off the revenue collected on this larger base against the investment lost through a marginal tax increase.

Profit shifting changes the after-tax profits firms receive from host investment when the government revises its tax. Without profit shifting, these profits are zero because the government taxes the entire host return. With profit shifting, firms can retain positive profits, so they invest even when they know that the government cannot commit. A better reputation then protects only the difference between the returns under commitment and discretion. I derive conditions under which the profits firms retain under discretion are large enough to make the optimal τ^R decrease with reputation. Equation (7) expresses the condition in terms of these profits and the investment response.

To establish this result analytically, let profits be $\rho(k) = zk^\alpha - r^*k$, with $z > 0$, $0 < \alpha < 1$, and $r^* \geq 0$, and let profit shifting costs be incurred and deducted only in the country from which profits are transferred. The host deducts profit shifting costs only on an outward transfer. With inward profit shifting its tax base is $\rho(K) + \Theta(K, \tau)$, and the cost is deducted abroad. Tax bases remain nonnegative. The timing, government types, and selection of a pooling equilibrium are unchanged.

Maintain the four parameter restrictions in (24), stated in Appendix A.4. The first condition ensures positive marginal profits and a declining elasticity of profits with respect to after-tax returns. The next two place the relevant taxes above the foreign rate and below the rate that exhausts the host tax base. The last imposes a lower bound on firms' after-tax returns under discretion. Appendices A.3 and A.4 derive the investment response and the lower bound on after-tax profits.

Proposition 1 (Lower optimal taxes with profit shifting). *Consider origin-country deductible profit shifting costs and $\rho(k) = zk^\alpha - r^*k$, with $z > 0$, $0 < \alpha < 1$, $r^* \geq 0$, and the four parameter restrictions in (24). Let $\tau^R(p)$ be any optimal tax under commitment selected as in Section 3.5. Then $\tau^* < \tau^R(p) < \tau^O$ for every $p \in (0, 1]$, and, for every $0 < p < p' \leq 1$,*

$$\tau^R(p') < \tau^R(p), \quad K(\tau^R(p'), p') > K(\tau^R(p), p).$$

These inequalities hold for every choice of maximizer at the two reputation levels.

Proof. See Appendix A.5. □

Proposition 1 places the optimal tax between the foreign and discretionary rates. Under (24), higher reputation increases the investment component of the tax-base semi-elasticity. The commitment type therefore chooses a lower rate. At the resulting allocation, firms invest more and shift a smaller fraction of their host profits abroad. The reported tax base expands sufficiently for revenue to increase.

Two further implications follow. First, the opportunistic type also earns more revenue at higher reputation because it inherits a larger investment base. Second, the

expected tax rate and expected outward profit shifting as a share of host profits both fall. The tax under commitment is lower and is implemented more often. Both types earn at least the revenue available after an off-path tax announcement, so their incentive compatibility constraints also hold.

After-tax profits under discretion. The tax comparison can be expressed without imposing quadratic profit shifting costs. Let $A(a)$ be the share of host profits accruing to the firm after taxes and profit shifting costs when the host tax rate is $a > \tau^*$. Write $A_O = A(\tau^O)$ for this share under discretion. The expected share is $w = pA(a) + (1 - p)A_O$. Investment $\mathcal{K}(w)$ equates $w\rho'(\mathcal{K})$ with $(1 - \tau^*)\rho'(\bar{k} - \mathcal{K})$. Define the elasticity of host profits with respect to this expected share and the rate at which the elasticity declines by

$$\varepsilon(w) = \frac{d \log \rho(\mathcal{K}(w))}{d \log w}, \quad \chi(w) = -\frac{d \log \varepsilon(w)}{d \log w} > 0.$$

At an interior optimum, the direct proportional revenue gain $1/a$ equals the semi-elasticity of the reported tax base with respect to the announced rate. The reporting component is independent of p . Reputation affects the investment component, $-\partial \log \rho(\mathcal{K}(w))/\partial a = -pA'(a)\varepsilon(w)/w$. Appendix A.6 shows that this component increases with reputation precisely when

$$A_O > p\chi(w)[A(a) - A_O]. \quad (7)$$

The left side is the share of profits firms receive under discretion. On the right, the additional share protected by commitment raises investment and reduces the elasticity of host profits. If (7) holds throughout the candidate tax interval, the revenue-maximizing announced rate falls with reputation. The quadratic restrictions in (24) ensure this inequality. Proposition A.1 establishes the same ordering and a unique optimal announced tax rate for a class of strictly convex profit shifting costs deductible in the country of origin, including $c(y) = \gamma y^2/2 + \eta y^3/3$ under its stated restrictions. The mechanism therefore extends beyond quadratic costs. The next comparison removes profit shifting while retaining the same production technology. It isolates the protection against a tax revision that drives Proposition 1.

4.2 Reputation without Profit Shifting

Without profit shifting, installed capital and the profits it generates are both exposed to host taxation. The opportunistic type chooses $\tau^O = 1$. Firms receive positive after-tax host profits only when the government is the commitment type, so investment satisfies

$$p(1 - \tau^R)\rho'(K) = (1 - \tau^*)\rho'(\bar{k} - K). \quad (8)$$

Better reputation now protects returns that would otherwise be confiscated. Its effect on investment is sufficiently strong that the government chooses a higher tax under commitment.

Proposition 2 (Higher optimal taxes without profit shifting). *Suppose profit shifting is unavailable and $\rho(k) = zk^\alpha - r^*k$, with $z > 0$, $0 < \alpha < 1$, $r^* \geq 0$, and $\alpha z \bar{k}^{\alpha-1} > (2 - \alpha)r^*$. For every $p \in (0, 1]$, (6) has a unique maximizer $\tau^R(p) \in (0, 1)$. For every $0 < p < p' \leq 1$,*

$$\tau^R(p') > \tau^R(p), \quad K(\tau^R(p'), p') > K(\tau^R(p), p).$$

Proof. See Appendix A.8. □

Profit shifting reverses the tax comparison by changing investment under the opportunistic type. Without profit shifting, firms lose the entire host return when the government is the opportunistic type. An improvement in reputation then has a sufficiently large effect on investment that the commitment type prefers to raise its rate. With profit shifting, firms obtain part of the return even after a tax revision, and some investment would take place without commitment. Under the conditions of Proposition 1, the government instead lowers its rate because the proportional revenue loss through investment increases with reputation.

This comparison concerns the choice of τ^R . The expected tax rate without profit shifting, $p\tau^R + (1 - p)$, still falls with p as confiscation becomes less likely. An expectation over the two government types is distinct from either the chosen policy or a realized statutory rate.

4.3 Numerical Illustration

The tax response affects both where firms invest and where they report profits. I compute these joint choices in the baseline with deductible costs in both countries. I set $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, $\bar{k} = 10$, $\tau^* = 0.15$, and $\gamma = 1.25$ in $\rho(k) = zk^\alpha - r^*k$, holding these parameters fixed as reputation changes. The comparison without profit shifting uses the same production parameters. The parameter values illustrate the mechanism. Appendix B.3 describes the solution and verification procedures.

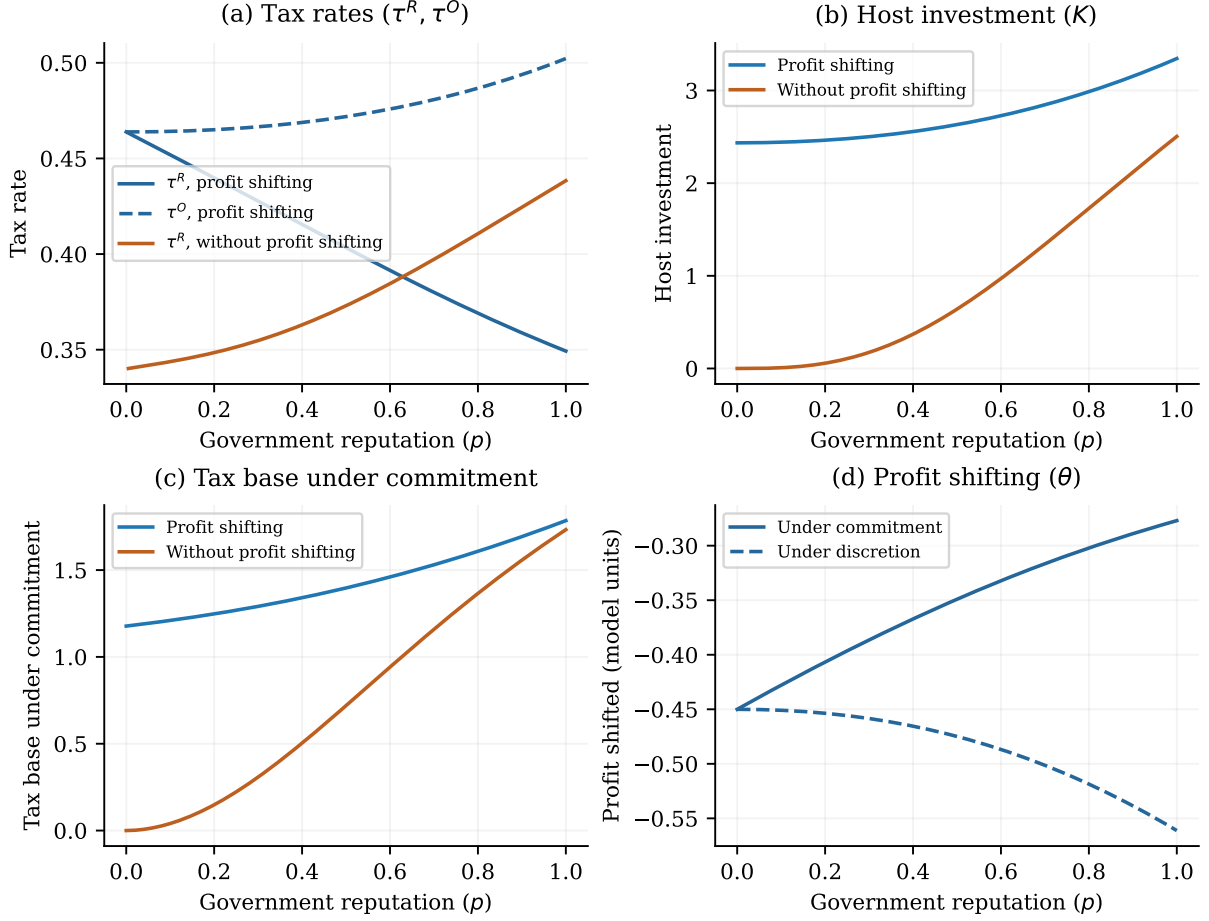


FIGURE 2: Reputation, tax policy, investment, and profit shifting

Notes: All panels use the baseline production parameters stated in the text. Profit shifting costs are deductible in both countries. Panel (a) distinguishes taxation under commitment from taxation under discretion. Panel (c) reports the tax base under commitment. Panel (d) reports profit shifting in model units, with negative values denoting outward profit shifting. The tax of the opportunistic type without profit shifting is one.

Figure 2 also gives an announced tax rate that declines with reputation. Increasing p from 0.1 to 0.9 lowers τ^R from 45.2 to 35.9 percent and raises K from 2.442 to 3.153. At the announced rate, the reported base expands from 1.210 to 1.693, while the outward transfer falls from 0.428 to 0.289. Hence revenue increases from 0.547 to 0.607 despite the reduction in the rate.

Under the opportunistic type, the additional capital is already installed when the tax is chosen. Its optimal rate rises from 46.4 to 49.4 percent, and firms respond by raising their outward transfer from 0.451 to 0.538. Thus higher reputation leads to less outward profit shifting if the government is the commitment type and more if it is the opportunistic type. The dependence of τ^O on capital reflects the costs incurred in both countries. Under origin-country costs, τ^O is constant.

Expected profit shifting averages the transfers across the two types. In the origin-country model, the expected outward share of host profits falls with p . The corresponding amount also reflects the increase in host profits as investment expands.

I repeat the tax comparison with non-deductible costs and with costs at the origin alone. I choose cost parameters to give the same reporting response at equal taxes and equal profits. The reference semi-elasticities are 1.26 and 1.50, guided by [Kallen \(2023\)](#) and [Beer et al. \(2018\)](#). All six comparisons give a tax under commitment that declines with reputation. Appendix [B.1](#) sets out the cost functions and reports the matched parameters and taxes.

5 Government Reputation and Future Investment

The one-period model gives the opportunistic type no reason to preserve its reputation after firms invest. I extend the model to two periods so that a tax revision affects future investment and revenue. The government can then prefer to implement its tax announcement even when a higher tax would increase current revenue. Firms anticipate this choice when investing. I first study the optimal announced tax rate when tax implementation is endogenous and compare the tax and investment responses with and without profit shifting. I then characterize policies implemented by both types and the investment sustained over longer horizons.

5.1 The Two-Period Model

A unit measure of firms enters in each period with the capital endowment and production technology in Section 3. Each firm operates for one period, and capital does not carry over between periods. Government type is fixed and privately known. The government maximizes current revenue plus second-period revenue discounted by $\beta \in (0, 1)$. Firms initially assign probability $p \in (0, 1)$ to the commitment type.

Within each period, the government announces a tax, firms invest, the government implements the tax, and firms choose their profit transfers. Second-period firms observe the first-period tax announcement and implemented tax before investing. Period 2 has no subsequent investment to protect, so its equilibrium is the one-period outcome selected in Section 3.5.

The analytical results use the convex origin-country costs in Proposition [A.1](#). Under these assumptions, the tax maximizing revenue on installed capital is τ^O , independent of capital. Write $\mathcal{R}(K, \tau)$ for revenue at installed capital K and implemented tax τ . For an outward transfer, $\mathcal{R}(K, \tau) = v(\tau)\rho(K)$, where v is revenue per unit of host profit. Let $V_C(q)$ and $V_O(q)$ denote the two types' revenues in the selected second-period

equilibrium at reputation q . Both equal R_0 at $q = 0$ and increase with q . Without profit shifting, the same notation applies with $\mathcal{R}(K, \tau) = \tau \rho(K)$, $\tau^O = 1$, and $R_0 = 0$.

5.2 Tax Implementation and Tax Announcements

Consider a common first-period tax announcement $a = \tau_1^R$. The commitment type implements it. Let m be the probability that the opportunistic type also implements it. For $a \neq \tau^O$, the probability that firms face the announced tax and the reputation after observing its implementation are

$$u = p + (1 - p)m, \quad q = \frac{p}{u}. \quad (9)$$

First-period investment is $K(a, u)$, using the investment function from the one-period model with u in place of p . Any tax revision reveals the opportunistic type and reduces reputation to zero. Conditional on a tax revision, that type chooses τ^O because every revised rate gives the same continuation reputation.

The opportunistic type compares the current gain from a tax revision with the discounted loss of future revenue. Define

$$D(a, m; p) = \mathcal{R}(K(a, u), \tau^O) - \mathcal{R}(K(a, u), a) - \beta[V_O(q) - R_0]. \quad (10)$$

Certain implementation requires $D \leq 0$, certain tax revision requires $D \geq 0$, and implementation with probability strictly between zero and one requires $D = 0$. These conditions determine m jointly with investment and the posterior q . Thus implementation is an equilibrium decision. When $m = 1$, investment is $K(a, 1)$ and reputation remains p . When $m = 0$, investment is $K(a, p)$ and observing the announced tax identifies the commitment type, giving $q = 1$.

As in Section 3.5, I compare common tax announcements according to the commitment type's revenue, subject to both types' incentive compatibility constraints. Its problem is

$$\max_{a \in [0, 1]} \{ \mathcal{R}(K(a, u(a)), a) + \beta V_C(q(a)) \}, \quad (11)$$

where $u(a)$ and $q(a)$ incorporate the opportunistic type's implementation decision. Firms assign probability zero to commitment after an off-path tax announcement. Appendix A.9 gives both incentive compatibility constraints and shows that an optimal common tax announcement exists. For $a < \tau^O$, the implementation probability is unique. At $a = \tau^O$, both types implement the same rate and reputation remains p .

The continuation payoff matters for the choice of the announced tax. A tax implemented by both types leaves reputation unchanged. A tax announcement that the

opportunistic type sometimes revises raises the commitment type's reputation when it implements that tax. The commitment type weighs this increase in future revenue against its effect on current investment and revenue. Consequently, finding the best tax that both types implement is one part of (11).

5.3 Optimal Tax Announcements at Low Reputation

Profit shifting also affects the tax announcement chosen when implementation is endogenous. Write $\tau_1^R(p; \beta)$ for a solution of (11). The next result establishes its response to reputation near zero, allowing the opportunistic type to implement the tax announcement with probability below one.

Proposition 3 (Optimal tax announcements with endogenous implementation). *Maintain the assumptions of Proposition A.1, and suppose A and v are four times continuously differentiable near τ^O . For every fixed $\beta \in (0, 1)$, there are $\bar{p}_\beta > 0$ and $c_\beta > 0$ such that the optimal common tax announcement is unique and strictly decreasing in p on $(0, \bar{p}_\beta)$. It satisfies*

$$\tau_1^R(p; \beta) = \tau^O - c_\beta p + O(p^2).$$

The opportunistic type's implementation probability has a strictly positive limit, which can be below one. Investment converges to $K_0 > 0$.

Proof. See Appendix A.10. □

The return available under discretion is central to this result. It gives the government a positive investment base even when its reputation is arbitrarily small. Near the discretionary optimum, a small reduction in the announced tax raises firms' retained profits to first order, while the additional revenue available from a tax revision is second order. Future revenue affects how much the opportunistic type implements and how quickly the announced tax rate falls. It does not reverse the direction of the tax response sufficiently close to zero reputation. The result applies to the government's preferred tax announcement, including when implementation is partial, and concerns the low-reputation region.

5.4 Profit Shifting, Tax Policy, and Investment

Figure 3 compares optimal announced taxes and real activity with and without profit shifting, allowing the opportunistic type to choose whether to implement the tax announcement. Production parameters and the discount factor are the same across the two economies. Let $K_{j,F}$ be investment under full commitment in economy $j \in \{S, N\}$,

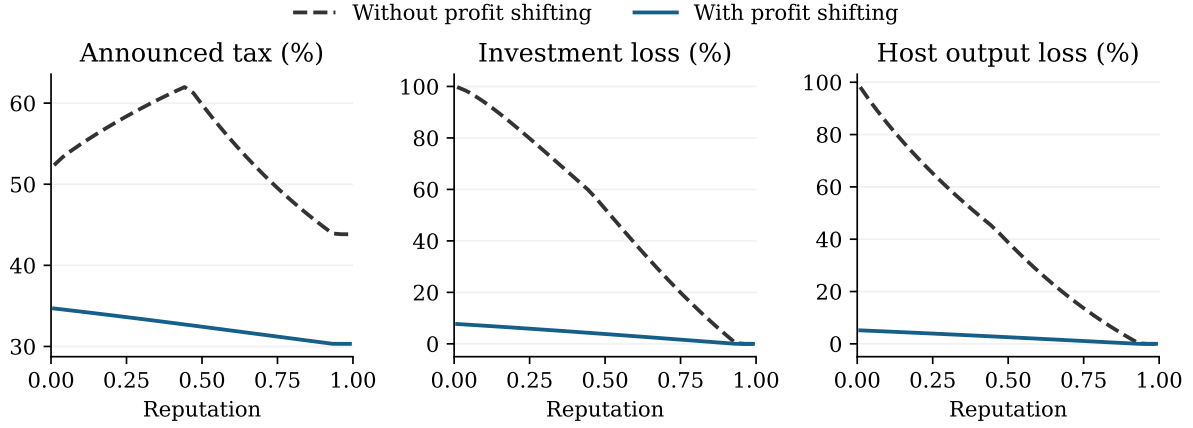


FIGURE 3: Optimal announced taxes and real activity under uncertain commitment

Notes: Each allocation maximizes the commitment type's discounted revenue among supported common tax announcements with endogenous implementation. The discount factor is 0.6. The economy with profit shifting uses origin-country deductible costs with $\gamma = 1.25$. Other parameters are those of Section 4.3. Investment and host output losses are measured against each economy's full-commitment allocation. Reputation is a model probability and is not fitted to the ICRG score.

where S permits profit shifting and N does not. The investment loss is $1 - K_j/K_{j,F}$, and the host output loss is $1 - (K_j/K_{j,F})^\alpha$. Each economy is compared with its own full-commitment allocation, since profit shifting also affects investment when commitment is certain.

Without profit shifting, panel (a) shows a nonmonotonic optimal announced tax. In the baseline comparison, the tax rises from 55.0 percent at $p = 0.1$ to 61.2 percent at $p = 0.4$ and then falls to 44.8 percent at $p = 0.9$. On the rising segment, the opportunistic type sometimes revises the tax announcement. Higher reputation and a higher implementation probability raise firms' expected returns and investment even as the announced rate increases. Once both types implement the preferred announced tax rate, a higher reputation increases the future revenue lost after a tax revision and allows the tax to fall toward its full-commitment level.

With profit shifting, the preferred announced tax rate is sustainable at every computed reputation and declines until the full-commitment tax is attained. Proposition 3 also establishes a declining preferred announced tax rate near zero reputation while allowing partial implementation. The numerical comparison thus preserves the direction of the static tax response when implementation is endogenous. Profit shifting gives a declining tax schedule consistent with the positive association between investment risk and statutory tax rates, whereas the economy without profit shifting has both rising and falling segments.

The investment differences are large at intermediate reputations. At $p = 0.5$, investment is 52.4 percent below full commitment without profit shifting and 3.8

percent below with profit shifting. The corresponding host output losses are 38.8 and 2.5 percent. At $p = 0.1$, the investment losses are 93.9 and 7.0 percent. These magnitudes illustrate the model rather than estimate country-specific losses. Appendix Table 4 also uses profit shifting costs matched to the reporting responses of 1.26 and 1.50 discussed in Appendix B.1. Both give smaller investment losses than the baseline economy with profit shifting, with the same production technology.

Without profit shifting, firms rely on the government's incentive to leave part of installed profits untaxed. Profit shifting provides an additional source of retained returns after the tax is implemented. The reporting response therefore affects how much real investment is lost under uncertain commitment. It need not make every announced tax easier to sustain, since the investment retained after a tax revision also reduces the subsequent revenue loss. The comparison concerns production in the host, while the capital endowment remains available for production abroad.

5.5 Sustainable Taxation and Investment

I first characterize the optimal tax conditional on implementation by both government types. I call such a tax sustainable. A tax revision raises current revenue but reveals the opportunistic type, reducing future investment. The tax implementation constraint is

$$\underbrace{\mathcal{R}(K(a, 1), \tau^O) - \mathcal{R}(K(a, 1), a)}_{G(a) \text{ current revenue gain}} \leq \beta \underbrace{[V_O(p) - R_0]}_{L(p) \text{ future revenue loss}}. \quad (12)$$

Both current revenues use the capital installed under the tax announcement. Firms can shift profits after a tax revision but cannot change that capital. The future revenue loss instead reflects the lower investment induced by learning that the government is the opportunistic type.

Within this set of tax announcements, $u = 1$ and $q = p$. Current investment does not depend on reputation at a fixed tax, and continuation revenue does not depend on the first-period tax announcement. Writing current revenue as $H(a) = \mathcal{R}(K(a, 1), a)$, the commitment type chooses

$$\max_{a \in [0, 1]} H(a) \quad \text{subject to } G(a) \leq \beta L(p) \text{ and (20)}. \quad (13)$$

Write $\tau_1^{R,s}(p)$ for this optimal sustainable tax, distinguishing it from the tax announcement chosen in (11).

Proposition 4 (Optimal sustainable taxes with profit shifting). *Maintain the assumptions of Proposition A.1 and fix $\beta \in (0, 1)$. The optimal sustainable tax $\tau_1^{R,s}(p)$ is unique. It equals the full-commitment tax $\tau^R(1)$ if that rate satisfies (12). Otherwise, it is the unique tax in $(\tau^R(1), \tau^O)$ at which the constraint binds. For every $0 < p < p' < 1$,*

$$\tau_1^{R,s}(p') \leq \tau_1^{R,s}(p), \quad \text{strictly if } \tau_1^{R,s}(p) > \tau^R(1).$$

Both incentive compatibility constraints hold at this tax.

Proof. See Appendix A.11. □

Raising the announced tax rate toward τ^O reduces the additional revenue available from a tax revision. If the full-commitment rate is not sustainable, a higher rate is required to satisfy the tax implementation constraint. Better reputation raises second-period investment following implementation and therefore increases $L(p)$. It allows a lower sustainable tax, which raises current investment and revenue until the full-commitment allocation is attained. Appendix A.15 gives the condition for attaining that allocation.

Future investment can also discipline taxation without profit shifting. The opportunistic type can then collect all current host profits by setting its tax to one, but doing so eliminates second-period investment.

Proposition 5 (Optimal sustainable taxes without profit shifting). *Suppose profit shifting is unavailable and maintain the production assumptions of Proposition 2. The optimal sustainable tax is unique. It equals $\tau^R(1)$ when that tax is sustainable and otherwise is the unique tax above it satisfying*

$$(1 - \tau_1^{R,s})\rho(K(\tau_1^{R,s}, 1)) = \beta\rho(K(\tau^R(p), p)). \quad (14)$$

As reputation rises, this tax weakly decreases and investment weakly increases, with strict changes while the tax exceeds $\tau^R(1)$. Both incentive compatibility constraints hold.

Proof. See Appendix A.12. □

Under certain implementation, reputation affects the sustainable tax through future revenue. Current firms face the announced tax under either government type, so their investment at a given rate is independent of reputation. Better reputation raises the future revenue lost after a tax revision and permits a lower current tax. This mechanism applies with or without profit shifting and characterizes $\tau_1^{R,s}$. The unrestricted choice τ_1^R in Section 5.4 also allows tax revisions and can therefore follow a different path.

Longer horizons. With more periods, the future revenue at stake includes returns from all remaining investment decisions. Following [Chari and Kehoe \(1990\)](#), a sustainable policy requires government policy and private decisions to be sequentially optimal after every policy history. Consider a finite horizon with implementation by both types before the terminal period and the selected one-period equilibrium at the terminal date. A tax revision reveals the opportunistic type and induces the discretionary allocation thereafter. At each earlier date, select the commitment type's preferred sustainable tax announcement given the continuation policy.

Let $E_h(p)$ denote the opportunistic type's revenue over h remaining periods in excess of the revenue following revelation. The terminal value is $E_1(p) = V_O(p) - R_0$. With $h \geq 2$ periods remaining, the sustainable tax $a_h(p)$ and this excess revenue satisfy

$$\begin{aligned} a_h(p) &= \arg \max_a H(a) \quad \text{subject to } G(a) \leq \beta E_{h-1}(p), \\ E_h(p) &= H(a_h(p)) - R_0 + \beta E_{h-1}(p). \end{aligned} \tag{15}$$

The tax $a_2(p)$ is $\tau_1^{R,s}(p)$ in the two-period model. The future cost of a tax revision now includes the revenue lost in all remaining periods.

Proposition 6 (Sustainable taxation over a finite horizon). *Maintain the assumptions of Proposition 4 or 5. For every fixed finite $h \geq 2$, (15) has a unique tax $a_h(p)$. This tax satisfies both types' incentive compatibility constraints and is weakly decreasing in reputation, strictly so whenever it exceeds $\tau^R(1)$. As $p \downarrow 0$, it converges to τ^O with profit shifting and to one without profit shifting. With profit shifting, investment, host production, and current revenue converge to $K_0 > 0$, $zK_0^\alpha > 0$, and $R_0 > 0$. Without profit shifting, all three converge to zero.*

Proof. See Appendix [A.13](#). □

These sustainable taxes approach their respective discretionary rates as reputation vanishes. With profit shifting, firms still retain a positive return at the discretionary tax and continue to invest. Without profit shifting, the limiting tax takes all host profits, so investment and production disappear. Profit shifting therefore determines whether sustainable taxation preserves a positive investment base in the low-reputation limit.

Investment at low reputation. The distinction in real activity extends to policies with endogenous implementation. Retain the environment with a common tax announcement and allow the opportunistic type to choose its implementation probability after every history. Continuation tax announcements may depend on the observed history and need only satisfy sequential incentives. Let $\mathcal{E}_h(p)$ denote these equilibria with h periods remaining and current reputation p . Write K_e for current investment in equilibrium e .

Proposition 7 (Investment with endogenous implementation over a finite horizon). *Fix any finite $h \geq 1$ and $\beta \in (0, 1)$. Under the production assumptions of Proposition 2, investment without profit shifting satisfies*

$$\lim_{p \downarrow 0} \sup_{e \in \mathcal{E}_h(p)} K_e = 0.$$

Current host production and tax revenue consequently converge to zero. With the origin-country profit shifting costs in Proposition A.1, there is a constant $\underline{K} > 0$ such that every equilibrium in $\mathcal{E}_h(p)$ satisfies $K_e \geq \underline{K}$, independently of p and h . The comparison does not depend on which common tax announcement the commitment type prefers.

Proof. See Appendix A.14. □

Without profit shifting, current investment relies on the prospect that the government leaves part of host profits untaxed. At low reputation, frequent implementation leaves little future reputation to preserve, while rare implementation provides little protection to current returns. This trade-off recurs in each remaining period, so investment approaches zero over any fixed finite horizon as reputation vanishes. Profit shifting preserves a positive return even after the opportunistic type is revealed. The reporting response therefore supports investment and production when reputation alone provides little protection.

6 Concluding Remarks

The relationship between government reputation and corporate taxation depends on firms' ability to protect returns after investment. The positive association between investment risk and statutory tax rates motivates a model in which profit shifting provides such protection. In the static model without profit shifting, better reputation protects otherwise confiscated profits, attracting enough investment for the government to announce a higher tax. With profit shifting, firms retain part of their returns under discretionary taxation. I show when this changes the investment response so that better reputation instead leads the government to announce a lower tax.

The dynamic analysis shows why profit shifting remains relevant to the relationship between reputation and statutory taxation. With profit shifting, I establish a declining optimal announced tax rate at low reputation. In the numerical comparison, the tax decreases toward its full-commitment level with profit shifting but first rises and then falls without profit shifting. This difference arises from the government's choice of the announced tax when implementation is endogenous. Profit shifting therefore connects

the model's tax response to the empirical risk-tax relationship as well as protecting real investment.

The implication for real activity is that firms need not rely entirely on the credibility of tax announcements. Over any fixed finite horizon, investment and host production approach zero as reputation approaches zero without profit shifting. With profit shifting, firms retain a positive return after taxation and continue to invest and produce even when government reputation provides little protection. The scope for profit shifting therefore matters for the real investment and production that an economy can sustain when tax-policy credibility is weak.

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A Analytical Results and Proofs

A.1 Profit Shifting and Investment

The notation follows Section 3. Aggregate profits and transfers are $\Lambda(K) = \rho(K)$ and $\Theta(K, \tau) = \theta(K, \tau)$ because firms have unit measure. Parameters not displayed as function arguments are held fixed.

Solving the two nonnegativity constraints on taxable profits gives

$$\begin{aligned} \theta &\geq \max \left\{ \frac{\rho(k)(1 - \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - k)(-1 - \sqrt{1 + 2\gamma})}{\gamma} \right\}, \\ \theta &\leq \min \left\{ \frac{\rho(k)(1 + \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - k)(-1 + \sqrt{1 + 2\gamma})}{\gamma} \right\}. \end{aligned} \quad (16)$$

The firm's payoff is strictly concave in the transfer. Its optimum is therefore the point in this interval closest to the unconstrained transfer in (3). At zero profits in either country, only a zero transfer is feasible. On the interior, substituting the optimal transfer gives host revenue

$$\tau \rho(K) \left\{ 1 + \frac{1}{2\gamma} - \frac{1}{2\gamma} \left[\frac{(1 - \tau^*)[\rho(K) + \rho(\bar{k} - K)]}{(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)} \right]^2 \right\}. \quad (17)$$

Its second derivative with respect to the tax is

$$\begin{aligned} & - \frac{\rho(K)(1 - \tau^*)^2[\rho(K) + \rho(\bar{k} - K)]^2\rho(\bar{k} - K)}{\gamma} \\ & \times \left\{ \frac{2}{[(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)]^3} + \frac{3\tau\rho(\bar{k} - K)}{[(1 - \tau)\rho(\bar{k} - K) + (1 - \tau^*)\rho(K)]^4} \right\} < 0. \end{aligned} \quad (18)$$

Thus revenue is strictly concave over the interval with nonbinding tax-base constraints. A binding foreign-base constraint gives a fixed positive host base and revenue increasing linearly with the tax. At the transition to the interior, the revenue derivative falls. A binding host-base constraint gives zero revenue, which cannot dominate a positive interior base. The government consequently has a unique tax best response at every positive interior capital allocation, including a possible maximum at one.

For investment, the function $-\theta^2/\rho(k)$ is jointly concave in positive profit and the transfer and increases with profit. Composition with the concave profit function preserves concavity. Both taxable bases are therefore concave in capital and the transfer, and requiring nonnegative bases gives a convex feasible set. The weighted sum in the firm's objective is strictly concave between distinct capital choices because $1 -$

$\tau^* > 0$ and foreign profits are strictly concave. Maximizing over transfers preserves strict concavity in capital. The individual investment optimum is unique for given implemented taxes. This does not establish uniqueness of the aggregate investment and tax fixed point.

At interior investment with nonbinding profit-shifting constraints, the first-order condition equates expected after-tax marginal profits,

$$\begin{aligned} \mathbb{E}_\tau \left[(1 - \tau) \left(1 + \frac{\gamma \theta(k, \tau)^2}{2\rho(k)^2} \right) \rho'(k) \right] \\ = (1 - \tau^*) \mathbb{E}_\tau \left[\left(1 + \frac{\gamma \theta(k, \tau)^2}{2\rho(\bar{k} - k)^2} \right) \rho'(\bar{k} - k) \right]. \end{aligned} \quad (19)$$

A.2 Equilibrium and Incentive Compatibility

An equilibrium with a government known to be the opportunistic type consists of mutual investment and tax best responses. With positive profit shifting costs, their pay-offs extend continuously to the compact capital and tax intervals under profit shifting subject to nonnegative tax bases. The individual investment best response is unique. The tax best response is unique at interior capital, all taxes are optimal at zero host capital, and a tax of one is optimal when all capital is in the host. Both correspondences have nonempty compact convex values and are upper hemicontinuous. Kakutani's theorem gives a fixed point, without establishing its uniqueness.

For the pooling equilibrium, firms retain prior p after the prescribed tax announcement. Each firm maximizes (5) while taking the implemented taxes as given. The commitment type implements its tax announcement and the opportunistic type solves (4). After an off-path tax announcement, beliefs assign probability one to the opportunistic type and investment is the capital in the chosen known-type equilibrium. The largest revenue either type could obtain after such a tax announcement is the revenue from its optimal tax on that capital. The two explicit inequalities in (6) require each type's equilibrium revenue to be at least this amount and prevent a deviation at the tax announcement stage. Firms optimize profit shifting after all subsequent histories. Bayes' rule does not restrict beliefs after a zero-probability tax announcement. These strategies and beliefs define a weak perfect Bayesian equilibrium. The off-path belief can be approached by fully supported trembles in tax announcements in which the commitment type's deviation probability vanishes faster than the opportunistic type's. The on-path posterior converges to the prior. The revenue selection in Section 3.5 ranks its supported common tax announcements.

When both types implement the first-period tax announcement, the continuation belief remains p after implementation and becomes zero after a tax revision. An off-path

tax announcement induces belief zero in both periods. Write $R_0 = R(0, \tau^O)$. Each type can obtain at most $(1 + \beta)R_0$ from such a tax announcement. The first-period tax must therefore satisfy

$$\begin{aligned} H(\tau_1^{R,s}) + \beta V_C(p) &\geq (1 + \beta)R_0, \\ H(\tau_1^{R,s}) + \beta V_O(p) &\geq (1 + \beta)R_0. \end{aligned} \quad (20)$$

These are the incentive compatibility constraints at the tax announcement stage, for the commitment type and the opportunistic type, respectively. The proofs of Propositions 4 and 5 verify these constraints together with the tax implementation constraint.

A.3 Net Profits and the Investment Response

Let $\rho(k) = zk^\alpha - r^*k$ and maintain $\alpha z\bar{k}^{\alpha-1} > (2 - \alpha)r^*$. Marginal profits are positive on $(0, \bar{k}]$. Differentiating the investment condition shows that the elasticity of host profits with respect to the ratio of after-tax returns is

$$\frac{\rho'(K)}{\rho(K) \left[-\frac{\rho''(K)}{\rho'(K)} - \frac{\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)} \right]}. \quad (21)$$

This expression decreases strictly with K . To see this, write its reciprocal as the sum of $-\rho(K)\rho''(K)/\rho'(K)^2$ and $[\rho(K)/\rho'(K)][-\rho''(\bar{k} - K)/\rho'(\bar{k} - K)]$. The first term increases when $r^* > 0$ and is constant when $r^* = 0$. In the second term, the first factor increases by concavity. The second factor increases with host capital because

$$-\frac{\frac{d}{dk} \left[-\frac{\rho''(k)}{\rho'(k)} \right]}{\left[-\frac{\rho''(k)}{\rho'(k)} \right]^2} = \frac{1 - (2 - \alpha)r^*/(\alpha z k^{\alpha-1})}{1 - \alpha} > 0. \quad (22)$$

The reciprocal of (21) therefore increases strictly. It tends to $(1 - \alpha)/\alpha$ as K tends to zero.

For $K \leq \bar{k}/2$, the rate at which this responsiveness decreases satisfies

$$\begin{aligned} &-\frac{1}{-\rho''(K)/\rho'(K) - \rho''(\bar{k} - K)/\rho'(\bar{k} - K)} \\ &\quad \times \frac{d}{dK} \log \left\{ \frac{\rho'(K)}{\rho(K) \left[-\rho''(K)/\rho'(K) - \rho''(\bar{k} - K)/\rho'(\bar{k} - K) \right]} \right\} \\ &\leq \frac{1}{1 - \alpha} \left[\frac{1}{2} + \frac{(2 - \alpha)r^*}{\alpha z (\bar{k}/2)^{\alpha-1}} \right]. \end{aligned} \quad (23)$$

To obtain the bound, use (22) and

$$\frac{\rho'(K)/\rho(K)}{-\rho''(K)/\rho'(K)} = \frac{\alpha}{1-\alpha} \frac{[1 - r^*/(\alpha z K^{\alpha-1})]^2}{1 - r^*/(z K^{\alpha-1})}.$$

Substitution into the logarithmic derivative bounds its left side in (23) above by

$$\begin{aligned} \frac{1}{1-\alpha} & \left\{ \frac{-\rho''(\bar{k}-K)/\rho'(\bar{k}-K)}{-\rho''(K)/\rho'(K) - \rho''(\bar{k}-K)/\rho'(\bar{k}-K)} \right. \\ & \left. + (2-\alpha) \left[\frac{-\rho''(K)/\rho'(K)}{-\rho''(K)/\rho'(K) - \rho''(\bar{k}-K)/\rho'(\bar{k}-K)} \right]^2 \frac{r^*}{\alpha z K^{\alpha-1}} \right\}. \end{aligned}$$

Equation (22) implies that host curvature is at least foreign curvature when $K \leq \bar{k}/2$. The first fraction is therefore at most one half, and the squared fraction is at most one. Finally, $K^{1-\alpha} \leq (\bar{k}/2)^{1-\alpha}$ gives (23). The same formulas imply that (21) is at most $\alpha/(1-\alpha)$.

A.4 Origin-Country Costs and After-Tax Returns

Suppose the parameters satisfy

$$\begin{aligned} \alpha z \bar{k}^{\alpha-1} & > (2-\alpha)r^*, \\ \gamma & > \frac{\tau^*}{1-\tau^*} + \frac{(\tau^*)^2}{2(1-\tau^*)^2}, \\ \frac{\tau^*}{\gamma(1-\tau^*)} \left(\frac{\sqrt{1+2\gamma}+1}{2} \right)^{\alpha/(1-\alpha)} & + \frac{\tau^*\alpha}{(1-\alpha)(1-\tau^*)} < 1, \\ \frac{\sqrt{1+2\gamma}-1}{2} \left[\frac{1}{2} + \frac{(2-\alpha)r^*}{\alpha z (\bar{k}/2)^{\alpha-1}} \right] & < 1-\alpha. \end{aligned} \tag{24}$$

With origin-country costs and $\rho(k) = zk^\alpha - r^*k$, an outward transfer has the interior solution

$$\frac{-\Theta(K, \tau)}{\Lambda(K)} = \frac{\tau - \tau^*}{\gamma(1-\tau)}, \quad 0 \leq \frac{-\Theta(K, \tau)}{\Lambda(K)} \leq \frac{\sqrt{1+2\gamma}-1}{\gamma}. \tag{25}$$

The upper bound follows from the nonnegative host base. Before it binds, firms retain total after-tax profits

$$\left[(1-\tau) + \frac{(\tau - \tau^*)^2}{2\gamma(1-\tau)} \right] \rho(K) + (1-\tau^*)\rho(\bar{k}-K). \tag{26}$$

The interior revenue-maximizing discretionary tax τ^O is independent of capital and solves

$$(2\gamma + 1)(1 - \tau^O)^3 = (1 - \tau^*)^2(1 + \tau^O). \quad (27)$$

The investment $K(\tau^O, 0)$ denotes equilibrium investment when firms know that this tax will be implemented. It is obtained from the marginal-profit condition

$$\left[(1 - \tau^O) + \frac{(\tau^O - \tau^*)^2}{2\gamma(1 - \tau^O)} \right] \rho'(K) = (1 - \tau^*)\rho'(\bar{k} - K). \quad (28)$$

Before the host-base constraint binds, the ratio of after-tax returns to host and foreign investment satisfies

$$\frac{(1 - \tau) + (\tau - \tau^*)^2 / [2\gamma(1 - \tau)]}{1 - \tau^*} \geq \frac{2}{\sqrt{1 + 2\gamma} + 1}. \quad (29)$$

To obtain this bound, differentiate the numerator with respect to the tax. The derivative is minus the host base per unit of profit, so the minimum occurs when this base reaches zero. Substituting the transfer at that point gives the right side. Once the host-base constraint binds, the transfer share remains $2/(\sqrt{1 + 2\gamma} + 1)$, the after-tax return to host investment equals $(1 - \tau^*)2/(\sqrt{1 + 2\gamma} + 1)$, and host revenue is zero. The same lower bound therefore holds up to a tax of one, using this constrained payoff rather than the interior formula.

The fourth restriction in (24) ensures that the after-tax return under discretion is large enough relative to the responsiveness bound in (23). The second and third restrictions exclude optimal taxes below τ^* as shown next.

A.5 Proof of Proposition 1

For an outward transfer, revenue per unit of host profits is

$$\tau \left[1 - \frac{\tau - \tau^*}{\gamma(1 - \tau)} - \frac{(\tau - \tau^*)^2}{2\gamma(1 - \tau)^2} \right]. \quad (30)$$

Its second derivative is $-(1 - \tau^*)^2(\tau + 2)/[\gamma(1 - \tau)^4] < 0$. Its first derivative at τ^* is $1 - \tau^*/[\gamma(1 - \tau^*)] > 0$. Revenue becomes zero when the host-base bound is reached. There is consequently a unique maximum above τ^* and below that bound, given by (27). The coefficient on host profits in the firm's after-tax return decreases over the positive-base interval. Its derivative is minus the host taxable base per unit of profit.

For $\tau < \tau^*$, inward profit shifting deducts its cost abroad and the transfer equals $(\tau^* - \tau)\rho(\bar{k} - K)/[\gamma(1 - \tau^*)]$. The second restriction in (24) makes this transfer feasible throughout the inward interval. Firm profits then equal

$$(1 - \tau)\rho(K) + \left[(1 - \tau^*) + \frac{(\tau^* - \tau)^2}{2\gamma(1 - \tau^*)} \right] \rho(\bar{k} - K).$$

The host return coefficient is at least the foreign coefficient. Under outward profit shifting, their ratio is at least $(\sqrt{1 + 2\gamma} - 1)/\gamma$. These bounds also hold under uncertainty about which tax is implemented. When $K \leq \bar{k} - K$, subtracting the rental cost gives $\rho'(K)/\rho'(\bar{k} - K) \geq [K/(\bar{k} - K)]^{\alpha-1}$ and $\rho(K)/\rho(\bar{k} - K) \geq [K/(\bar{k} - K)]^\alpha$. The investment condition and the lower bound on after-tax profits therefore imply

$$\frac{\rho(K)}{\rho(\bar{k} - K)} \geq \left(\frac{\sqrt{1 + 2\gamma} - 1}{\gamma} \right)^{\alpha/(1-\alpha)}.$$

If $K > \bar{k} - K$, the same lower bound holds because the profit ratio exceeds one. For fixed such capital, the derivative of revenue over tax rates inducing inward profit shifting is $\rho(K) + (\tau^* - 2\tau)\rho(\bar{k} - K)/[\gamma(1 - \tau^*)] > 0$ by the third restriction. No tax rate inducing inward profit shifting improves on τ^* . The global tax best response is therefore the same τ^O at every equilibrium capital allocation.

For announced tax rates between τ^* and τ^O , investment satisfies

$$\begin{aligned} & \left\{ p \left[(1 - \tau^R) + \frac{(\tau^R - \tau^*)^2}{2\gamma(1 - \tau^R)} \right] + (1 - p) \left[(1 - \tau^O) + \frac{(\tau^O - \tau^*)^2}{2\gamma(1 - \tau^O)} \right] \right\} \rho'(K) \\ & = (1 - \tau^*)\rho'(\bar{k} - K). \end{aligned} \quad (31)$$

Capital increases with p and decreases with the announced tax rate. An announced tax rate above τ^O reduces both the firm's after-tax return and the government's revenue per unit of profit, so it is dominated by τ^O .

To exclude announced tax rates below τ^* , differentiate the investment condition using the inward-transfer payoff. Its host coefficient is $p(1 - \tau^R) + (1 - p)[(1 - \tau^O) + (\tau^O - \tau^*)^2/(2\gamma(1 - \tau^O))]$ and its foreign coefficient is $(1 - \tau^*) + p(\tau^* - \tau^R)^2/[2\gamma(1 - \tau^*)]$. The second restriction implies that their ratio decreases with τ^R . Thus investment exceeds $K(\tau^O, 0)$ throughout this interval. Log differentiation also gives

$$\frac{\partial \log \rho(K)}{\partial \tau^R} \geq -\frac{\alpha}{(1 - \alpha)(1 - \tau^*)}.$$

Using the bound on the profit ratio, the derivative of revenue under the announced tax divided by host profits is at least

$$1 - \frac{\tau^*}{\gamma(1 - \tau^*)} \left(\frac{\sqrt{1 + 2\gamma} + 1}{2} \right)^{\alpha/(1-\alpha)} - \frac{\tau^* \alpha}{(1 - \alpha)(1 - \tau^*)} > 0,$$

where the strict inequality is the third restriction in (24). Revenue therefore increases below τ^* and immediately to its right. At τ^O , its derivative is negative when $p > 0$, since the direct revenue derivative is zero and capital falls with the announced tax rate. All optimal announced tax rates lie strictly between the two rates.

It remains to order these interior maxima. Log differentiation of revenue under the announced tax gives the derivative of the log of (30), which is independent of p , plus the derivative of the after-tax profit coefficient, which is negative, multiplied by

$$\frac{p\rho'(K)^2}{(1 - \tau^*)\rho'(\bar{k} - K)\rho(K) \left[-\frac{\rho''(K)}{\rho'(K)} - \frac{\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)} \right]}. \quad (32)$$

The bound in (23) implies that the derivative of this factor with respect to p , after division by positive terms, is bounded below by

$$(1 - \tau^O) + \frac{(\tau^O - \tau^*)^2}{2\gamma(1 - \tau^O)} - \frac{1}{1 - \alpha} \left[\frac{1}{2} + \frac{(2 - \alpha)r^*}{\alpha z(\bar{k}/2)^{\alpha-1}} \right] \left[(1 - \tau^*) - (1 - \tau^O) - \frac{(\tau^O - \tau^*)^2}{2\gamma(1 - \tau^O)} \right] > 0. \quad (33)$$

To obtain this bound, differentiate the product of p , the investment responsiveness in (21), and the reciprocal of the expected after-tax profit coefficient. The direct effect of p contributes the after-tax return under discretion. The capital response contributes at worst minus p times the bound in (23) times the difference between the commitment and discretionary after-tax returns. That difference is at most the foreign after-tax return minus the discretionary after-tax return. The fourth restriction in (24) and (29) make the displayed lower bound positive. Log revenue consequently has strictly decreasing differences in the announced tax rate and reputation. Adding the two maximization inequalities excludes a higher announced tax rate at higher reputation. Equality is excluded by the interior first-order conditions, since the log-revenue derivative falls strictly with reputation. This orders every optimal announced tax rate without assuming uniqueness.

A higher p and a lower announced tax rate both raise investment. Revenue under the announced tax rises strictly because it rises with p at each fixed announced tax rate below τ^O . Dividing this higher revenue by the lower tax gives a larger tax base

under the announced tax. The opportunistic type's revenue also rises because (30) at τ^O is a fixed positive coefficient multiplying host profits. Announcing τ^O reproduces the known-type allocation at any prior. The selected tax announcement yields at least that revenue under the announced tax and more investment, so both types' incentive compatibility constraints hold. The stated off-path beliefs support these allocations for $0 < p < 1$. At $p = 1$, the selected tax announcement is the full-commitment optimum.

Finally, the tax and outward share under commitment are below their counterparts under discretion and decrease with reputation. The expected tax rate and expected outward share,

$$p\tau^R + (1-p)\tau^O, \quad -\frac{p\Theta(K, \tau^R) + (1-p)\Theta(K, \tau^O)}{\Lambda(K)}, \quad (34)$$

therefore decrease strictly. The outward share in (25) does not depend on capital. The amount shifted additionally depends on the expanding host profits.

A.6 After-Tax Profits and General Profit Shifting Costs

This appendix extends the reputation comparative static in Proposition 1. Costs remain deductible at the origin, and all timing and information assumptions are unchanged. If the outward transfer is a share y of host profits, let its cost be $c(y)\rho(K)$. An inward transfer of share x of foreign profits costs $c(x)\rho(\bar{k} - K)$. Suppose c is three times continuously differentiable, $c(0) = c'(0) = 0$, and $c'' > 0$. Define $B = 1 - \tau^*$ and let $\bar{y} \in (0, 1)$ solve $1 - \bar{y} - c(\bar{y}) = 0$.

For an outward transfer of share y , write $b(y) = 1 - y - c(y)$. The share of host profits accruing to the firm after taxes and profit shifting costs is

$$A(\tau) = \max_{0 \leq y \leq \bar{y}} \{(1 - \tau)b(y) + By\}.$$

An interior transfer satisfies $\tau(y) = 1 - B/[1 + c'(y)]$. This tax increases strictly with y . The envelope theorem gives $A'(\tau) = -b(y(\tau))$ until the host base vanishes, after which $A(\tau) = B\bar{y}$. Let $v(\tau) = \tau b(y(\tau))$.

Proposition A.1 (Reputation under general origin-country costs). *Maintain the production specification and first restriction in (24). Put $\kappa = \min_{[0, \bar{y}]} c''(y) > 0$ and*

$$C_\rho = \frac{1}{1-\alpha} \left[\frac{1}{2} + \frac{(2-\alpha)r^*}{\alpha z(\bar{k}/2)^{\alpha-1}} \right].$$

Suppose

$$\begin{aligned} c'(\bar{y}) &> \frac{\tau^*}{B}, \\ \frac{\tau^*}{B\kappa} \bar{y}^{-\alpha/(1-\alpha)} + \frac{\tau^* \alpha}{(1-\alpha)B} &< 1, \\ C_\rho \left(\frac{1}{\bar{y}} - 1 \right) &< 1, \\ c'''(y)[1 + c'(y)] &\leq 2[c''(y)]^2 \quad \text{for } 0 \leq y \leq \bar{y}. \end{aligned} \tag{35}$$

The unique discretionary tax τ^O is independent of capital at every investment allocation induced by a tax announcement. The commitment type's optimal tax announcement is unique and satisfies $\tau^ < \tau^R(p) < \tau^O$. For every $0 < p < p' \leq 1$,*

$$\tau^R(p') < \tau^R(p), \quad K(\tau^R(p'), p') > K(\tau^R(p), p).$$

Both types' revenues increase strictly, and both incentive compatibility constraints hold. The expected tax rate and the expected outward transfer share decrease strictly.

Proof. Put $h = 1 + c'$. Revenue per unit of profit, as a function of the outward share, is $g(y) = \tau(y)b(y)$. Direct differentiation gives

$$\begin{aligned} g'(y) &= \frac{Bc''b}{h^2} - \tau h, \\ g''(y) &= \frac{B[c'''h - 2(c'')^2]b}{h^3} - \frac{2Bc''}{h} - \tau c'' < 0. \end{aligned} \tag{36}$$

The second condition in (35) implies $B\kappa > \tau^*$, so $g'(0) > 0$. At $\bar{y}$, $g'(\bar{y}) < 0$. Thus g has a unique interior maximum, corresponding to $\tau^O > \tau^*$. The coefficient $v(\tau)$ increases before τ^O and decreases after it until the base vanishes.

For a tax $a < \tau^*$ inducing inward profit shifting, the first condition makes the reporting choice interior, with $c'(x) = (\tau^* - a)/B$ and $x < \bar{y}$. The after-tax coefficients on host and foreign profits are $1 - a$ and

$$F(a) = B + (\tau^* - a)x - Bc(x).$$

Their difference is $(\tau^* - a)(1 - x) + Bc(x) \geq 0$. At every tax rate inducing outward profit shifting the ratio of these coefficients is at least $\bar{y}$. The ratio in any probability-weighted combination is consequently at least $\bar{y}$. The investment condition and the argument in Appendix A.5 then give

$$\frac{\rho(K)}{\rho(\bar{k} - K)} \geq \bar{y}^{\alpha/(1-\alpha)}.$$

At installed capital satisfying this bound, the derivative of revenue over tax rates inducing inward profit shifting is

$$\rho(K) + \left[x - \frac{a}{Bc''(x)} \right] \rho(\bar{k} - K) > 0.$$

The second condition therefore excludes tax rates inducing inward profit shifting from the global optimum under discretion. The same τ^O is the optimal discretionary tax at every relevant capital allocation. Write $A_O = A(\tau^O)$. The envelope derivative and the interiority of τ^O imply

$$A_O > B\bar{y}, \quad A(\tau^*) = B. \quad (37)$$

For a tax announcement $a < \tau^*$, the expected coefficients are $U = p(1 - a) + (1 - p)A_O$ and $V = pF(a) + (1 - p)B$. Since $F'(a) = -x$,

$$-\frac{1}{1 - a} \leq \frac{\partial}{\partial a} \log \frac{U}{V} = -\frac{p}{U} + \frac{px}{V} \leq 0.$$

The last inequality follows from $x(1 - a) \leq F(a)$, equivalent to $x + c(x) \leq 1$, and $xA_O \leq B$. The profit elasticity in (21) is bounded by $\alpha/(1 - \alpha)$. Differentiating the commitment type's inward revenue and dropping its nonnegative terms gives

$$\frac{1}{\rho(K)} \frac{d}{da} \{a[\rho(K) + x\rho(\bar{k} - K)]\} \geq 1 - \frac{\tau^* \alpha}{(1 - \alpha)B} - \frac{\tau^*}{B\kappa} \bar{y}^{-\alpha/(1-\alpha)} > 0.$$

The right derivative at τ^* is also positive. An announced tax rate above τ^O lowers both A and v , so it is dominated by τ^O . At τ^O , the derivative of the commitment type's revenue is negative for $p > 0$, because $v' = 0$ and investment declines with the announced tax rate. Every optimal announced tax rate is consequently interior to (τ^*, τ^O) .

For these tax announcements, put $w = pA(a) + (1 - p)A_O$. Investment $\mathcal{K}(w)$ solves $w\rho'(\mathcal{K}) = B\rho'(\bar{k} - \mathcal{K})$. Define

$$\varepsilon(w) = \frac{d \log \rho(\mathcal{K}(w))}{d \log w}, \quad \chi(w) = -\frac{d \log \varepsilon(w)}{d \log w}. \quad (38)$$

These quantities depend on production and investment, not on the reporting function at a fixed w . Appendix A.3 shows $\varepsilon > 0$, $\chi > 0$, and $\chi \leq C_\rho$ when $K \leq \bar{k}/2$. Since $w \leq B$ in the outward profit-shifting region, that bound applies here.

Log revenue satisfies the exact identities

$$\begin{aligned}\frac{\partial \log R}{\partial a} &= \frac{v'(a)}{v(a)} + A'(a) \frac{p\varepsilon(w)}{w}, \\ \frac{\partial^2 \log R}{\partial p \partial a} &= \frac{A'(a)\varepsilon(w)}{w^2} \{A_O - p\chi(w)[A(a) - A_O]\}.\end{aligned}\quad (39)$$

The expression in braces is bounded below by $A_O - C_\rho(B - A_O) > 0$, using (37) and the third condition. Thus log revenue has strictly decreasing differences in the announced tax rate and reputation. The two optimality inequalities exclude a higher tax at a higher prior. A common interior maximizer is excluded by (39).

To establish uniqueness, put $q(a) = b(y(a))$, so $v = aq$ and $A' = -q$. Interior reporting gives

$$q' = -\frac{h^3}{Bc''} < 0, \quad q'' = -\frac{h^4}{B^2c''} \left[3 - \frac{hc'''}{(c'')^2} \right] < 0.$$

At any stationary point of revenue with respect to the announced tax, let $s = v'/v = pq\varepsilon/w$. Substituting this first-order condition into the second derivative gives

$$\frac{\partial^2 \log R}{\partial a^2} = \frac{q' + aq'' - a(q')^2/q}{aq} - s^2 \left[1 + \frac{1 + \chi(w)}{\varepsilon(w)} \right] < 0. \quad (40)$$

Every stationary point is therefore a strict maximum. The positive derivative at τ^* and negative derivative at τ^O imply that exactly one exists. It is the unique global maximizer because announced tax rates below the foreign rate and announced tax rates above τ^O have already been excluded. The strict second derivative also gives a differentiable optimum for positive priors.

Both a higher prior and a lower announced tax rate increase w and investment. At each fixed announced tax rate below τ^O , reputation raises revenue under the announced tax, so its maximum rises strictly. The opportunistic type's revenue is the fixed coefficient $v(\tau^O)$ times the higher host profits. Announcing τ^O reproduces the known-type allocation, and each selected tax announcement induces more investment. Both incentive compatibility constraints therefore hold. Finally, the announced tax and its outward transfer share are below their discretionary counterparts and decline with p . Their probability-weighted averages decline as well. $\square$

The cost class includes $c(y) = \gamma y^2/2 + \eta y^3/3$ with $0 \leq \eta \leq \gamma^2$, provided the remaining restrictions in (35) hold. Here $\kappa = \gamma$, and the curvature condition reduces to $\eta \leq \gamma^2 + 3\gamma\eta y + 3\eta^2 y^2$. At $\eta = 0$, the conditions reduce to the quadratic restrictions in (24). Strictness therefore gives a nonempty neighborhood of nonquadratic costs around every admissible quadratic specification. At the baseline parameters, $\eta = 0.25, 0.5, 0.75$ satisfy all restrictions. The same cost class gives the sustainable tax comparison in Proposition 4 and the full-commitment threshold in Proposition A.2.

Equation (39) also identifies the role of the after-tax return. At any differentiable strict local maximum, the sign of the tax response is the opposite of the expression in braces. If profit shifting is unavailable, $A_O = 0$ and $A(a) = 1 - a$. The braces are then negative because $\chi > 0$, giving the increasing tax response in Proposition 2. With a sufficiently large after-tax return under discretion, they are positive and the response is reversed.

A.7 Choice of the Announced Tax

This appendix verifies the selection of the tax announcement in Section 3.5 for pure tax announcements under Proposition A.1 and in the benchmark without profit shifting.

Write $R_0 = v(\tau^O)\rho(K(\tau^O, 0))$. Either type can obtain R_0 by announcing τ^O , regardless of firms' belief, because the implemented tax is then τ^O for both types. In a separating tax announcement equilibrium, the opportunistic type is identified before investment and receives R_0 . If the commitment type announces $a < \tau^O$, firms invest more than $K(\tau^O, 0)$. The opportunistic type can therefore gain by imitating this tax announcement and subsequently imposing τ^O . If $a > \tau^O$, the commitment type receives less than R_0 , since both the host investment return and revenue per unit of host profit are lower. Its own incentive compatibility constraint rules this out. Thus a separating tax announcement outcome can only have $a = \tau^O$ and give both types R_0 . By Proposition A.1, both types receive strictly more at the selected common tax announcement. Separation at the tax announcement stage is therefore dominated in equilibrium payoffs. Without profit shifting, the same argument applies with $\tau^O = 1$ and $R_0 = 0$, while the selected common tax announcement gives both types positive revenue.

For completeness, Lu's strongly coherent criterion selects this common tax announcement. Among the types attributed to an alternative tax announcement, a strictly improving type sends with probability one, an indifferent type may send, and a type that loses cannot be included. Beliefs must be consistent with those incentives. Consider an equilibrium with a common tax announcement that does not maximize the commitment type's revenue. The maximizing common tax announcement is off path

and belongs to an alternative equilibrium in which that type strictly gains. A belief consistent with the improving types assigns probability at least p to commitment. Since revenue at the optimal announced tax rate increases with that belief, the gain remains positive and rules out the candidate. For a separating candidate, both types gain from a common tax announcement near the optimum that is outside its path, giving the same conclusion.

Conversely, no alternative pure tax announcement equilibrium improves the commitment type's payoff at the selected optimum. If an alternative benefits a type, it can only be the opportunistic type. Belief zero after an off-path tax announcement is therefore consistent with Lu's criterion and supports the selected allocation. This verifies the selection for the one-period problem and hence the terminal-period continuation used in Section 5.³ Separation through an initial tax announcement is distinct from revealing the opportunistic type through a subsequent tax revision. The two-period model therefore determines the implementation decision separately from the initial common tax announcement.

A.8 Proof of Proposition 2

Without profit shifting, (8) implies $K_{\tau^R} < 0$ and $K_p > 0$. Differentiating the government's objective $\tau^R \rho(K)$ gives

$$\tau^R(p) = 1 - \left\{ 1 + \left[-\frac{\rho''(K)}{\rho'(K)} - \frac{\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)} \right] \frac{\rho(K)}{\rho'(K)} \right\}^{-1}. \quad (41)$$

By Appendix A.3, the expression in braces increases strictly with K . Substituting this tax into

$$p = \frac{1 - \tau^* \rho'(\bar{k} - K)}{1 - \tau^R \rho'(K)}$$

gives a strictly increasing function of capital. It tends to zero as K tends to zero and to infinity as K tends to $\bar{k}$. There is exactly one stationary tax for every $p > 0$, and both capital and this tax increase with p . Revenue is positive at interior taxes and zero at zero and one, so the unique stationary tax is the global maximizer. A government known to be the opportunistic type attracts no capital and earns no revenue. Each type's positive revenue at the selected tax announcement therefore satisfies its incentive compatibility constraint. Higher capital in (8) also requires a higher after-tax profit coefficient $p(1 - \tau^R)$. The expected tax rate $1 - p(1 - \tau^R)$ therefore decreases.

More generally, this argument applies to any three times continuously differentiable profit function with positive marginal profits, strict concavity, an infinite marginal profit

³Lu's Section 5 explains why the intuitive criterion of [Cho and Kreps \(1987\)](#) can be unsuitable when a tax announcement binds only the commitment type.

at zero, and the expression in braces in (41) that is strictly increasing and bounded near zero. The rental-cost specification satisfies these requirements under the condition stated in Proposition 2.

A.9 Tax Implementation and the Choice of the Announced Tax

Fix $p \in (0, 1)$ and $\beta \in (0, 1)$, and retain the terminal selection and off-path beliefs in Section 5.2. For $a < d = \tau^O$, investment increases with the implementation probability m . On the outward profit-shifting interval, the current gain from a tax revision is $[v(d) - v(a)]\rho(K(a, u))$, which therefore increases with m . For $a < \tau^*$, let $x(a)$ denote the inward transfer per unit of foreign profit. The gain is

$$[v(d) - a]\rho(K(a, u)) - ax(a)\rho(\bar{k} - K(a, u)).$$

It also increases with investment since $v(d) > \tau^* > a$. Without profit shifting, the gain is $(1 - a)\rho(K(a, u))$ and has the same property. Meanwhile $q = p/[p + (1 - p)m]$ decreases with m , and $V_O(q)$ increases with q . Thus $D(a, m; p)$ is strictly increasing in m for every $a < d$.

There is consequently one implementation probability at each such tax announcement. It is zero if $D(a, 0; p) \geq 0$, one if $D(a, 1; p) \leq 0$, and otherwise the unique interior root of $D(a, m; p) = 0$. This probability is continuous in a . At $a = d$, both types implement d and the posterior is p . Since $V_O(p) > R_0$, announced tax rates sufficiently close to d from below have $m = 1$, making the allocation continuous at this endpoint as well.

The commitment type's and the opportunistic type's incentive compatibility constraints at the tax announcement stage are

$$\begin{aligned} \mathcal{R}(K(a, u), a) + \beta V_C(q) &\geq (1 + \beta)R_0, \\ m[\mathcal{R}(K(a, u), a) + \beta V_O(q)] + (1 - m)[\mathcal{R}(K(a, u), d) + \beta R_0] &\geq (1 + \beta)R_0. \end{aligned} \quad (42)$$

At $m = 1$, these reduce to (20). The tax d satisfies both, so supported common tax announcements exist. The constraints define a closed set, and the commitment type's objective is continuous on $[0, d]$.

Announced tax rates above d cannot improve its optimum. To see this, fix any supported tax announcement $a > d$ and its implementation probability. Its investment is at most K_0 , its current revenue is below R_0 , and the discretionary revenue on this investment is at most R_0 . If $m = 1$, announcing d gives higher current revenue and the same continuation. If $m < 1$, the current gain from a tax revision is at least $\beta[V_O(q) - R_0]$, with equality when $m > 0$. Holding u fixed, the gain from a tax revision decreases continuously from $v(d)\rho(K(0, u))$ to zero as the announced tax rate rises from zero to d .

Its initial value is at least the gain under $a > d$. There is therefore a tax announcement $\tilde{a} < d$ whose gain equals $\beta[V_O(q) - R_0]$. The same m is consistent with implementation incentives at $\tilde{a}$, and investment is at least K_0 . The commitment type's current revenue equals discretionary revenue on this larger investment minus $\beta[V_O(q) - R_0]$, exceeding its revenue at a . Its continuation is unchanged. The opportunistic type's payoff also weakly increases, so both incentive compatibility constraints remain satisfied. This establishes existence of an optimal common tax announcement on $[0, d]$. It does not require uniqueness of the optimal announced tax rate or a ranking of all dynamic equilibria.

Without profit shifting, a low weight on future revenue gives a useful analytical comparison. Let $I = [p_L, p_H] \subset (0, 1)$ and write $t(p) = \tau^R(p)$ for the selected one-period tax. There is $\bar{\beta}_I > 0$ such that, for every $p \in I$ and $0 < \beta < \bar{\beta}_I$, the commitment type's preferred common tax announcement is $t(p)$ and the opportunistic type revises it with certainty.

To prove this statement, write $S(w) = \rho(\mathcal{K}(w))$, where $w\rho'(\mathcal{K}) = (1 - \tau^*)\rho'(\bar{k} - \mathcal{K})$. Put $P(q) = S(q[1 - t(q)])$ and $C(q) = t(q)P(q)$. These functions are increasing, continuous, and zero at $q = 0$. If $m = 0$, the commitment type obtains $aS(p[1 - a]) + \beta C(1)$, uniquely maximized at $t(p)$. The gain from a tax revision at this tax announcement, $(1 - t(p))P(p)$, has a positive minimum on I , so $m = 0$ is optimal for the opportunistic type when β is sufficiently small.

It remains to compare tax announcements with $m > 0$. Their implementation incentive requires $(1 - a)S(u[1 - a]) \leq \beta P(q) \leq \beta P(1)$. Let $M = \min_{p \in I} C(p) > 0$ and choose $\delta > 0$ so that $S(\delta) < M/3$. If $1 - a < \delta$, current revenue is less than $M/3$. Otherwise it is at most $\beta P(1)/\delta$, which is less than $M/3$ for sufficiently small β . Continuation revenue is at most $\beta C(1)$ and can also be made less than $M/3$. Every such tax announcement then gives less than $2M/3$, while $t(p)$ with $m = 0$ gives at least M . Both incentive compatibility constraints hold because the off-path payoff is zero. Proposition 2 implies that this optimal announced tax rate increases with reputation.

A.10 Proof of Proposition 3

Put $d = \tau^O$, $v_0 = v(d)$, $A_0 = A(d)$, and $S(w) = \rho(\mathcal{K}(w))$. All derivatives in this proof are evaluated on the outward profit-shifting interval. Define

$$\begin{aligned} S_0 &= S(A_0) > 0, & S_1 &= S'(A_0) > 0, \\ b &= -A'(d) > 0, & h &= -v''(d) > 0, & \kappa &= \frac{v_0 b S_1}{h S_0} > 0. \end{aligned}$$

The strict curvature follows from (36) and the strictly increasing map from the transfer share to the tax. The stated smoothness gives smooth local investment and revenue functions. The terminal tax announcement $t(q)$ solves

$$v'(t)S(w) + v(t)S'(w)qA'(t) = 0, \quad w = A_0 + q[A(t) - A_0].$$

At $(t, q) = (d, 0)$, the derivative of this equation with respect to t is $-hS_0$. The implicit-function theorem yields $t(q) = d - \kappa q + O(q^2)$. Hence, writing $L(q) = V_O(q) - R_0$ and $J(q) = V_O(q) - V_C(q)$,

$$L(q) = \ell q^2 + O(q^3), \quad J(q) = \frac{1}{2}\ell q^2 + O(q^3), \quad \ell = hS_0\kappa^2 > 0. \quad (43)$$

Indeed, $w - A_0 = b\kappa q^2 + O(q^3)$ and $v_0 - v(t) = h\kappa^2 q^2/2 + O(q^3)$. The identity defining κ gives the stated coefficients. Also $J(q) > 0$ for every $q > 0$, because $t(q) < d$ and the opportunistic type collects strictly more revenue on the same installed capital.

First consider the location of an optimal announced tax rate as $p \downarrow 0$. Any sequence with posterior $q \geq \delta > 0$ after implementation has $u = p/q \rightarrow 0$. Investment then converges uniformly to K_0 , including for tax announcements inducing inward transfers. If $0 < m < 1$, the tax implementation constraint gives the commitment type the value

$$v_0\rho(K) + \beta R_0 - \beta J(q). \quad (44)$$

If $m = 0$, its value is at most this expression with $q = 1$. Thus such sequences yield strictly less than $(1 + \beta)R_0$ in the limit. Announcing d yields $R_0 + \beta V_C(p) \rightarrow (1 + \beta)R_0$. Every optimizing sequence must therefore have $q \rightarrow 0$ or certain implementation. In either case, the tax implementation constraint and $L(q) \rightarrow 0$ imply $a \rightarrow d$: investment is bounded away from zero, and the gain from a tax revision is bounded away from zero for announced tax rates bounded away from d . Announced tax rates above d are dominated by the argument in Appendix A.9. Consequently, for small p , an optimal announced tax rate lies in the outward profit-shifting region near d , and $m = 0$ is excluded.

For partial implementation, use q to parameterize the binding tax implementation constraint, with $u = p/q$ and $q \geq p$. Set $x = d - a$. The constraint is

$$[v_0 - v(d - x)]S\left(A_0 + \frac{p}{q}[A(d - x) - A_0]\right) = \beta L(q). \quad (45)$$

There is a smooth solution $x = qZ(p, q)$ near $(p, q) = (0, 0)$ with

$$Z(0, 0) = \kappa\sqrt{2\beta} > 0.$$

To see this, substitute $x = qZ$ into (45) and divide by q^2 . Both sides extend smoothly to $q = 0$. The equation at the origin is $hS_0Z^2/2 = \beta\ell$, and its derivative with respect to the positive root Z is nonzero. The terms $[A(d - qZ) - A_0]/q$ also have a smooth extension. The implicit-function theorem therefore applies. The endpoint $q = p$ gives certain implementation at the binding sustainable tax. For small p that tax is above the full-commitment rate, so it is the best tax announcement among those implemented with certainty.

Using (44), define the commitment type's value along this curve by

$$U(p, q) = v_0S \left(A_0 + \frac{p}{q}[A(d - qZ(p, q)) - A_0] \right) - \beta J(q) + \beta R_0.$$

The smooth extension satisfies

$$\begin{aligned} U(0, 0) &= (1 + \beta)R_0, & U_q(0, 0) &= 0, \\ U_p(0, 0) &= v_0S_1b\kappa\sqrt{2\beta} > 0, \\ U_{qq}(0, 0) &= -\beta\ell < 0. \end{aligned}$$

In particular, U is strictly concave in q in a neighborhood of the origin. The unconstrained maximizer there is smooth and satisfies

$$\hat{q}(p) = \lambda_\beta p + O(p^2), \quad \lambda_\beta = \frac{U_{pq}(0, 0)}{\beta\ell}.$$

The constrained choice is $q(p) = \max\{p, \hat{q}(p)\}$. The preceding localization argument excludes all remaining tax announcements for sufficiently small p . Both incentive compatibility constraints hold at this choice: the commitment type's value and the opportunistic type's value exceed $(1 + \beta)R_0$ by $v_0S_1b\kappa\sqrt{2\beta}p + O(p^2)$. Thus the unconstrained comparison does not select an unsupported tax announcement.

It follows that

$$c_\beta = \kappa\sqrt{2\beta}\max\{1, \lambda_\beta\} > 0, \quad \lim_{p \downarrow 0} m(p) = \frac{1}{\max\{1, \lambda_\beta\}}. \quad (46)$$

On each smooth branch the tax derivative converges to $-c_\beta$. If $\lambda_\beta = 1$, both branches have the same strictly negative limiting derivative, and their constrained selection remains strictly decreasing. Strict concavity in q and uniqueness of the best certain-implementation tax establish uniqueness of the tax announcement. Since the expected retained share converges to A_0 , investment converges to K_0 .

Let $c_A = A''(d)$ and $j = v'''(d)$. Expanding (45) one order further gives

$$\lambda_\beta = \kappa \left[\frac{3c_A/(2b) - j/(2h)}{\sqrt{2\beta}} + \frac{c_A}{b} - \frac{j}{3h} \right].$$

This expression determines the limiting implementation probability. No restriction on its sign is needed for the declining tax, since (46) uses $\max\{1, \lambda_\beta\}$.

A.11 Proof of Proposition 4

Maintain the assumptions of Proposition A.1, and use its functions A , v , and $\mathcal{K}$. Write $S(w) = \rho(\mathcal{K}(w))$, $d = \tau^O$, $v_O = v(d)$, and $R_0 = v_O S(A_O)$. The proposition establishes a unique discretionary tax, a unique full-commitment tax $\tau^F = \tau^R(1) \in (\tau^*, d)$, and strictly increasing continuation revenues for both types.

With certain implementation of $a \in [\tau^*, d]$, current revenue and the gain from a tax revision are

$$H(a) = v(a)S(A(a)), \quad G(a) = [v_O - v(a)]S(A(a)). \quad (47)$$

Investment is strictly decreasing in a because $A' < 0$ and $\mathcal{K}' > 0$. The uniqueness proof in Appendix A.6, evaluated at $p = 1$, establishes that H increases up to τ^F and decreases thereafter. On $[\tau^*, d]$, $v_O - v(a) > 0$ and $v'(a) > 0$, giving

$$G'(a) = -v'(a)S(A(a)) + [v_O - v(a)]S'(A(a))A'(a) < 0.$$

Thus G decreases continuously to zero at d .

For $a < \tau^*$, firms shift profits into the host. Appendix A.6 establishes that investment decreases and current revenue increases with a throughout this interval. In particular, the host-to-foreign return ratio $(1 - a)/F(a)$ decreases strictly, since the inward transfer satisfies $x + c(x) < 1$. Discretionary revenue $v_O \rho(K(a, 1))$ therefore decreases with a , so G decreases strictly below τ^* as well. Above d , current revenue is below R_0 because both the return to host investment and revenue per unit of host profits are lower, or the host tax base is zero.

Fix $p \in (0, 1)$. Both continuation revenues are at least R_0 , and $L(p) = V_O(p) - R_0 > 0$. Any first-period tax yielding at least R_0 therefore satisfies both incentive compatibility constraints. The rate d does so and has $G(d) = 0$. The full-commitment rate yields strictly more than R_0 . If $G(\tau^F) \leq \beta L(p)$, it is feasible and uniquely optimal. Otherwise, the strict decrease of G gives a unique $a \in (\tau^F, d)$ with $G(a) = \beta L(p)$. Every lower rate violates the tax implementation constraint. Every higher rate up to d yields less current

revenue, and rates above d yield less than R_0 . The binding rate is therefore uniquely optimal and satisfies both incentive compatibility constraints.

Increasing p raises $L(p)$ while leaving $G(a)$ unchanged at any fixed announced tax rate. The binding rate decreases strictly until it reaches τ^F , at which point further increases in reputation have no effect. Each strict tax reduction raises investment. Increasing β has the same effect because it raises the discounted continuation loss. These arguments use the properties of the general cost class rather than a quadratic specification.

A.12 Proof of Proposition 5

Without profit shifting, a government known to be the opportunistic type attracts zero capital and earns zero revenue. Under implementation with certainty of a tax $a < 1$, investment $K(a, 1)$ is positive and strictly decreasing in a . The revenue $H(a) = a\rho(K(a, 1))$ is zero at both endpoints and has the unique interior maximum $\tau^F = \tau^R(1)$ established in Appendix A.8. It increases before that tax and decreases afterward. The gain from a tax revision to one is

$$G(a) = (1 - a)\rho(K(a, 1)).$$

It is strictly decreasing on $[0, 1)$ because both positive factors decrease, and it converges to zero at one.

The opportunistic type's terminal revenue after implementing the first-period tax announcement is $\rho(K(\tau^R(p), p))$. Proposition 2 makes it strictly positive and strictly increasing in p , even though the terminal tax chosen by the commitment type also increases. Terminal revenue following revelation is zero. If $G(\tau^F)$ does not exceed the discounted loss of future tax revenue, the full-commitment tax is feasible and is the unique optimum. Otherwise, the strictly decreasing G has exactly one root of (14) above τ^F . Lower taxes violate the tax implementation constraint and higher taxes reduce current revenue. Increasing p raises the right side, lowering the binding tax and increasing investment. Each type earns positive current and continuation revenue, while an off-path tax announcement gives zero, so both incentive compatibility constraints hold.

A.13 Proof of Proposition 6

For any continuation excess revenue $E \geq 0$, the proof of Proposition 4, or its counterpart without profit shifting, gives a unique sustainable optimum $a(E)$. It equals τ^F if $G(\tau^F) \leq \beta E$ and otherwise solves $G(a(E)) = \beta E$ on $(\tau^F, \tau^O]$. Thus $a(E)$ is continuous

and nonincreasing, strictly decreasing whenever it exceeds τ^F . The maximized current revenue $H(a(E))$ is continuous and nondecreasing in E , and is at least R_0 .

The mapping $F(E) = H(a(E)) - R_0 + \beta E$ is continuous and strictly increasing, with $F(0) = 0$. The terminal excess revenue $E_1(p) = V_O(p) - R_0$ is continuous, strictly increasing in p , and converges to zero as $p \downarrow 0$. Applying F repeatedly establishes these properties for every finite E_h . It follows that $a_h(p) = a(E_{h-1}(p))$ decreases with reputation while its tax implementation constraint binds. Continuity at zero gives $a_h(p) \rightarrow \tau^O$. Investment consequently converges to $K_0 > 0$ with profit shifting and to zero without it, where $\tau^O = 1$ in the latter case. Continuity of $Y_H = zK^\alpha$ and of current revenue then gives the output and revenue limits.

The revenue after revelation over h periods is $R_0 \sum_{j=0}^{h-1} \beta^j$. Current sustainable revenue is at least R_0 , and both types' terminal revenues are at least R_0 . Backward induction therefore gives each type at least its payoff from an off-path tax announcement. The opportunistic type's tax implementation constraint is exactly the first line of (15). Following a tax revision or an off-path tax announcement, the known-type discretionary allocation is sequentially optimal in each remaining period. Firms optimize at all histories. These conditions support the recursively selected policy as a sustainable allocation within the maintained class.

For the comparison across horizons, the same increasing map implies that $E_2(p) \geq E_1(p)$ makes the entire sequence $E_h(p)$ nondecreasing and $a_h(p)$ nonincreasing in h . If the initial inequality is reversed, the subsequent ordering is also reversed. This condition accounts for the fact that the opportunistic type's terminal revenue is collected at the discretionary tax, whereas earlier sustainable revenue is collected at the announced tax. More periods alone do not establish a larger continuation loss under every parameter configuration.

A.14 Proof of Proposition 7

First consider the economy without profit shifting. Write $\mathcal{K}(w)$ for investment when the expected coefficient on host profits is w and the foreign coefficient is $B = 1 - \tau^*$. Thus $w\rho'(\mathcal{K}(w)) = B\rho'(\bar{k} - \mathcal{K}(w))$. Set $S(w) = \rho(\mathcal{K}(w))$, with $\mathcal{K}(0) = S(0) = 0$. Both functions are continuous and strictly increasing on $[0, 1]$. Current revenue is at most $S(1)$, so the opportunistic type's discounted revenue over h periods is bounded by $M_h = S(1) \sum_{j=0}^{h-1} \beta^j$. Once the opportunistic type is revealed, backward induction gives confiscation and zero investment at every remaining date, hence zero continuation revenue.

Let $B_h(\delta)$ be the supremum of the opportunistic type's discounted revenue over all sequentially rational continuations with h periods remaining, common tax announce-

ments, and beliefs in $[0, \delta]$. The supremum includes different choices of the announced tax and histories. I prove by induction that $B_h(\delta) \rightarrow 0$ as $\delta \downarrow 0$ and that current investment converges to zero uniformly over these continuations. In the terminal period, any tax announcement $a < 1$ is revised with certainty by the opportunistic type. The expected retained share is $w = p(1 - a) \leq p$, giving $B_1(\delta) \leq S(\delta)$. The tax announcement $a = 1$ also gives zero investment.

Suppose the claims hold for $h - 1$. At a current tax announcement $a < 1$, let m be the opportunistic type's implementation probability, $u = p + (1 - p)m$, $q = p/u$, and $w = u(1 - a)$. If investment did not converge uniformly to zero, a sequence with $p \downarrow 0$ would have $w \geq \epsilon > 0$. Then $u \geq \epsilon$ and $1 - a \geq \epsilon$, so $m > 0$ for sufficiently small p . Implementation must then be weakly preferred to confiscation. Consequently,

$$\epsilon S(\epsilon) \leq (1 - a)S(w) \leq \beta B_{h-1}(p/\epsilon),$$

where $q \leq p/\epsilon$. The right side converges to zero by the induction hypothesis, a contradiction. The tax announcement $a = 1$ gives zero current investment directly. Current investment and revenue therefore converge to zero uniformly.

It remains to bound expected continuation revenue conditional on the opportunistic type, because implementation can occasionally lead to a high posterior. For any $\eta \in (0, 1)$, let $W_{h-1}(q)$ denote its continuation value after implementation. If $q \leq \eta$, this value is bounded by $B_{h-1}(\eta)$. If $q > \eta$, Bayes' rule gives $u = p/q < p/\eta$ and $m \leq u/(1 - p)$. Thus

$$mW_{h-1}(q) \leq B_{h-1}(\eta) + \frac{p}{\eta(1 - p)}M_{h-1}.$$

Following a tax revision the continuation value is zero. First letting $p \downarrow 0$ and then $\eta \downarrow 0$ proves that expected continuation revenue also vanishes uniformly. At $a = 1$, the implemented tax conveys no information and the continuation value is at most $\beta B_{h-1}(p)$. This completes the induction. The production and current-revenue limits follow from $Y_H = zK^\alpha$ and $0 \leq \mathcal{R}(K, \tau) \leq \rho(K)$.

With profit shifting, let $\bar{y} \in (0, 1)$ solve $1 - \bar{y} - c(\bar{y}) = 0$. The ratio of optimized after-tax coefficients on host and foreign profits is at least $\bar{y}$ at every implemented tax. For an outward transfer this follows from $A(\tau) \geq B\bar{y}$. For an inward transfer it follows from $(1 - \tau) \geq F(\tau)$ in Appendix A.6. The bound is preserved by every probability-weighted combination of implemented taxes. The investment condition therefore gives

$$K \geq \underline{K} > 0, \quad \bar{y}\rho'(\underline{K}) = \rho'(\bar{k} - \underline{K}).$$

This bound is independent of reputation, implementation incentives, and the number of remaining periods. It also gives $Y_H \geq z\underline{K}^\alpha > 0$. No selection among supported common tax announcements is used in either part of the argument.

A.15 Implementation of the Full-Commitment Tax

Let $\tau^F = \tau^R(1)$, $K_F = K(\tau^F, 1)$, and K_0 denote investment under a government known to be the opportunistic type. Write v_O and v_F for revenue per unit of host profit under discretion and at τ^F . Without profit shifting, $v_O = 1$, $v_F = \tau^F$, and $K_0 = 0$.

Proposition A.2 (When reputation sustains the full-commitment tax). *Maintain the assumptions of Proposition 4 or 5, and define*

$$\beta_{\min} = \frac{(v_O - v_F)\rho(K_F)}{v_O[\rho(K_F) - \rho(K_0)]} \in (0, 1). \quad (48)$$

If $\beta \leq \beta_{\min}$, no reputation $p \in (0, 1)$ supports implementation of the full-commitment first-period tax by both types. If $\beta > \beta_{\min}$, there is a unique $p_F \in (0, 1)$ such that both types implement this tax exactly when $p \geq p_F$. The threshold p_F decreases strictly with β . Without profit shifting, $\beta_{\min} = 1 - \tau^F$.

Proof. In either economy, the undiscounted loss of future tax revenue is

$$\Delta(p) = v_O\{\rho(K(\tau^R(p), p)) - \rho(K_0)\}.$$

It is continuous and strictly increasing on $(0, 1)$, tends to zero as $p \downarrow 0$, and tends to $v_O[\rho(K_F) - \rho(K_0)]$ as $p \uparrow 1$. The limits without profit shifting follow from Appendix A.8. In the origin-country benchmark, continuity of the investment problem and the unique endpoint optima give the limits. The uniqueness result in Proposition A.1, which includes the quadratic benchmark, gives continuity of the interior optimal announced tax rate by the maximum theorem.

The first-period gain at τ^F is $G_F = (v_O - v_F)\rho(K_F) > 0$. Moreover, $v_F\rho(K_F) > v_O\rho(K_0)$, because the full-commitment optimum strictly improves on the known-type allocation. Therefore $0 < G_F < \Delta(1)$, where $\Delta(1)$ denotes the limit. Their ratio is (48). If $\beta \leq G_F/\Delta(1)$, the strict inequality $\Delta(p) < \Delta(1)$ rules out implementation at every $p < 1$. If $\beta > G_F/\Delta(1)$, continuity and strict monotonicity give exactly one root of $\beta\Delta(p_F) = G_F$. The full-commitment tax is feasible precisely above that root. Increasing β lowers the root strictly. Its current revenue exceeds the known-type revenue, so both incentive compatibility constraints hold whenever the tax implementation constraint is satisfied. Without profit shifting, substitution of $v_O = 1$, $v_F = \tau^F$, and $K_0 = 0$ gives the last claim.

□

B Numerical Methods and Robustness

This appendix checks whether the static tax comparison depends on the treatment of profit shifting costs and describes the numerical verification of the main results. The calculations distinguish the tax announcement problem with endogenous implementation from the optimal sustainable tax. The final subsection gives the solution method for the bilateral cost specification.

B.1 Cost Treatment and Reporting Responses

The cost functions used here apply to transfers inside an existing multinational firm. Establishing the multinational structure itself would involve a different cost. This distinction appears in the structural estimates of [Bilicka et al. \(2024\)](#) and [Altshuler et al. \(2025\)](#). Operational costs, audit exposure, and financial-reporting incentives are discussed in [McClure \(2023\)](#). I vary the location and deductibility of the resource costs, choosing their parameters to give the same reporting semi-elasticity at a specified reference allocation.

I match the response of the reported tax base to the host tax rate, holding capital fixed. At equal tax rates and equal profits, the negative derivative of the log host base is $1/[2\gamma(1 - \tau^*)]$ under bilateral deductions. The baseline $\gamma = 1.25$ and $\tau^* = 0.15$ imply a semi-elasticity of 0.471. The additional reference values are 1.26 from [Kallen \(2023\)](#) and 1.50 from [Beer et al. \(2018\)](#). A value of 1.50 also corresponds to the short-run aggregate estimate in [Bilicka et al. \(2024\)](#), which holds capital and tax-planning capacity fixed. Matching these responses makes the alternative cost specifications comparable at a common allocation. It does not identify a structural value of γ from the estimates, whose populations and tax rates differ from that allocation.

With non-deductible costs in both countries, the firm's reporting objective is

$$(1 - \tau)[\rho(k) + \theta] + (1 - \tau^*)[\rho(\bar{k} - k) - \theta] - \frac{\gamma\theta^2}{2} \left[\frac{1}{\rho(k)} + \frac{1}{\rho(\bar{k} - k)} \right]. \quad (49)$$

The unconstrained transfer is $(\tau^* - \tau)\rho(k)\rho(\bar{k} - k)/[\gamma(\rho(k) + \rho(\bar{k} - k))]$. Imposing nonnegative bases gives the constrained transfer, and the reference semi-elasticity is $1/(2\gamma)$. For costs deducted only at the origin, (25) instead gives $1/[\gamma(1 - \tau^*)]$ on the outward profit-shifting side. Costs of inward profit shifting are deducted abroad. The government's maximization is carried out over the entire tax interval in each specification. I compare interior optima with the endpoint taxes zero and one and with the rates at which a tax-base constraint binds.

TABLE 3: Optimal taxes with alternative profit shifting costs

Costs incurred in	Tax deductible?	Tax-base semi-elasticity	Tax under commitment (%)		
			$p = 0.1$	$p = 0.5$	$p = 0.9$
Both countries	Yes	1.26	31.5	29.3	27.2
Both countries	No	1.26	38.1	33.8	30.2
Country from which profits are shifted	Yes	1.26	30.6	29.3	27.9
Both countries	Yes	1.50	29.2	27.3	25.6
Both countries	No	1.50	34.3	30.9	28.0
Country from which profits are shifted	Yes	1.50	28.5	27.3	26.2

Notes: Taxes are in percent. Each reference response holds capital fixed at equal tax rates and equal profits before profit shifting. Production parameters are the same within each comparison. Nonnegative tax bases apply throughout. The rows match responses at this reference allocation, not at every equilibrium.

TABLE 4: Real activity relative to full commitment

Reporting response	Investment loss (%)			Host output loss (%)		
	$p = 0.1$	$p = 0.5$	$p = 0.9$	$p = 0.1$	$p = 0.5$	$p = 0.9$
Without profit shifting	93.91	52.44	3.51	84.22	38.77	2.33
Baseline	7.05	3.82	0.30	4.71	2.54	0.20
1.26	4.96	2.68	0.20	3.30	1.77	0.13
1.50	4.05	2.18	0.16	2.69	1.44	0.11

Notes: Two-period optimal common tax announcements with endogenous implementation and $\beta = 0.6$. Losses are relative to each economy's own full-commitment allocation. The baseline origin-country cost has $\gamma = 1.25$ and reporting semi-elasticity $1/[\gamma(1 - \tau^*)] = 0.941$ at the reference allocation. The other rows with profit shifting set $\gamma = 1/[e(1 - \tau^*)]$ for reporting response e . Production parameters are unchanged. All reported allocations with profit shifting have certain implementation. The allocation without profit shifting has partial implementation at $p = 0.1$ and certain implementation at $p = 0.5$ and 0.9 .

Table 3 compares costs incurred in both countries with costs incurred only at the origin, with and without tax deductibility. Each of the six comparisons gives a lower optimal tax under commitment at a higher reputation. Matching the reporting response at a common allocation thus preserves the direction of the baseline result across these specifications.

B.2 General Costs and the Comparison without Profit Shifting

For the general-cost result in Proposition A.1, the numerical specification is $c(y) = \gamma y^2/2 + \eta y^3/3$, with $\gamma = 1.25$ and $\eta = 0, 0.25, 0.5, 0.75$. The remaining parameters are those of Section 4.3. Each specification satisfies the strict inequalities in (35). I first find the unique discretionary optimum on the outward profit-shifting interval, then solve the investment condition jointly with the first-order condition for the commitment type's tax announcement. The seven priors used are $p = 0.01, 0.1, 0.25, 0.5, 0.75, 0.9, 0.99$. In all four specifications, τ^R is decreasing and K increasing across these priors, as

predicted by the proposition. Solving the problem without profit shifting at the same values of p gives the increasing sequence of announced rates in Proposition 2.

For Proposition 4, I compute sustainable taxes for all four cost functions at $p = 0.01, 0.1, 0.5, 0.9, 0.99$ and $\beta = 0.3, 0.6, 0.9$. Each of the 60 allocations satisfies the investment condition and both incentive compatibility constraints. Taxes decrease with reputation until the full-commitment rate becomes feasible. I compare each choice with 1,001 alternative tax rates and the regime boundaries, allowing inward transfers and binding tax-base constraints. The maximum investment residual is below 1.2×10^{-16} . For quadratic costs and the economy without profit shifting, separate calculations using the seven priors above compare tax choices with 2,001 points in $[0, 1]$. Two independent solutions for quadratic costs differ in their announced rates by less than 6×10^{-11} .

The local tax expansion in Proposition 3 is also checked in 55-digit arithmetic. I compare its tax slope and limiting implementation probability with solutions of the binding tax implementation constraint and the commitment type's first-order condition. These checks include both partial and certain implementation. The finite-reputation investment comparisons retain the incentive compatibility constraints and compare a finer tax search at the reported reputations. These computations supplement the local analytical result and do not establish an all-reputation tax theorem.

B.3 Numerical Solution and Verification

I solve the investment and discretionary-tax problems jointly at each announced tax rate, then maximize the commitment type's revenue over tax announcements that satisfy both types' incentives. Individual firms take the implemented tax as given. To find the best discretionary tax, I compare revenue at zero and one, at every transition into a binding tax-base constraint, and at interior maxima between transitions.

For Figure 3 and Table 4, I solve the implementation conditions in (10) at each announced tax rate, update reputation by Bayes' rule, and evaluate both types' discounted revenues. The search includes the boundaries between certain tax revision, partial implementation, and certain implementation, the foreign tax rate, and the tax endpoints. I refine every local maximum identified on a grid of 121 announced tax rates and repeat the comparison with 1,001 announced tax rates at the three reputations reported in the table. The two searches agree in the commitment type's value to within 10^{-9} . Both incentive compatibility constraints and the relevant implementation condition hold at every reported allocation. An independent comparison over 2,001 implemented rates holds capital fixed when checking the tax revision. Appendix A.9 excludes announced tax rates above τ^O analytically. These numerical checks concern

the maintained problem with a common tax announcement, not uniqueness among all dynamic equilibria.

For sustainable taxes, investment is computed under certain implementation, and I maximize current revenue subject to (12) and (20). For longer horizons, I apply (15) recursively using the same terminal values.

I check the reported allocations with separate maximizations over investment, reporting, and implemented taxes, and use a finer search over announced tax rates to check the policy choice. I also verify that the quadratic cost specification reproduces the baseline payoff function.

At $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, $\bar{k} = 10$, and $\tau^* = 0.15$, all four restrictions in (24) hold for $\gamma = 0.5$ and 1.25. The baseline parameters also satisfy the restriction without profit shifting in Appendix A.8.

C Data and Additional Empirical Results

This appendix describes the construction of investment risk and reports the controlled regressions and sample comparisons used in the text.

C.1 Sources and Estimation Samples

The risk series comes from the Investment Profile sheet of ICRG Table 3B, January 1984–December 2021. I require twelve reported monthly scores to calculate an annual mean and subtract that mean from twelve. The three underlying subcomponent scores are not separately available in this file. Historical country aggregates are not assigned to successor countries. I merge the annual risk measure with tax rates dated one year later.

The OECD combined statutory rate gives 1,907 complete annual observations in 101 countries during 2003–2021 before the source exclusions below. The main estimation sample contains 1,887 observations in the same countries. Economic controls are real GDP per capita in constant 2015 US dollars, population, and natural-resource rents/GDP from the World Bank. Country coverage is determined by OECD rates and the controls, without requiring observations for other outcomes.

The additional ICRG controls use the same twelve-month requirement and one-year lag as Investment Profile. I subtract corruption and law and order scores from six, bureaucratic quality from four, and government stability and internal and external conflict from twelve. These are the component maxima in the ICRG methodology. All

six controls are observed throughout the tax estimation sample, so adding them does not change its composition.

The downloaded OECD file records zero rates for Haiti in 2003–2010 and Togo in 2003–2014. Historical IMF reports describe positive corporate tax rates during these intervals. For Haiti, the 2006 reform replaced a graduated schedule with a 30 percent proportional rate ([International Monetary Fund, 2007](#)). For Togo, IMF documentation reports positive rates before and after the 2009–2010 reductions ([International Monetary Fund, 2010](#)). I exclude these inconsistent zero records from the main estimation sample, without imputing replacement rates, and retain the original-record estimates as a source comparison. This removes twenty observations, leaving 1,887 observations in 101 countries. Excluding both countries entirely leaves 1,869 observations in 99 countries. Other zero-rate jurisdictions remain in the sample.

C.2 Tax Regressions and Source Checks

Annual regressions cluster by country, with the number of countries minus one as the reference degrees of freedom. Table 5 reports all slope coefficients for columns (4) and (5) of Table 1. Table 6 keeps the full set of controls fixed and compares the original OECD records with the two source-check samples. Reported tests are specification-specific and do not adjust for multiple testing.

Adding lagged GDP growth to the fully controlled annual regression gives coefficients of 0.354, 0.420, and 0.424 in the recorded, main, and country-exclusion samples, respectively, with p -values of 0.020, 0.005, and 0.004. With economic controls alone, the original-record coefficient is 0.263 ($p = 0.148$), compared with 0.395 ($p = 0.009$) after the source exclusions. The original records therefore support the positive association most precisely when political risks are also controlled for.

Leaving out one country at a time gives fully controlled annual coefficients between 0.280 and 0.396 in the recorded sample. All are positive, and 100 of the 101 estimates are significant at 5 percent. Omitting Liberia gives $p = 0.053$. After excluding the questionable zero records, the coefficients range from 0.371 to 0.457, and every country exclusion gives $p < 0.009$. Thus no single country's observations account for the conditional association in the latter sample.

TABLE 5: Within-country tax regressions with full control coefficients

	(4)	(5)
Investment risk	0.426 (0.143)	0.420 (0.145)
Log real GDP per capita	1.653 (1.337)	1.708 (1.354)
Log population	4.364 (1.894)	4.390 (1.887)
Resource rents/GDP	0.041 (0.033)	0.043 (0.033)
Corruption risk	-0.417 (0.306)	-0.427 (0.305)
Law and order risk	0.013 (0.471)	0.011 (0.473)
Bureaucracy risk	1.419 (1.029)	1.419 (1.028)
Government stability risk	0.102 (0.114)	0.098 (0.114)
Internal conflict risk	-0.336 (0.239)	-0.334 (0.240)
External conflict risk	-0.080 (0.311)	-0.088 (0.310)
Lagged GDP growth	–	-0.016 (0.024)
Country and year effects	Yes	Yes
Observations	1,887	1,887

Notes: Columns (4) and (5) of Table 1. All ICRG measures are lagged by one year and expressed as risks. Income and population are logged, and resource rents/GDP and statutory rates are in percentage points. Parentheses contain country-clustered standard errors. Dashes indicate controls omitted from a specification.

TABLE 6: OECD tax regressions and the treatment of questionable zero records

	(1)	(2)	(3)
Investment risk	0.353	0.426	0.430
Standard error	(0.150)	(0.143)	(0.143)
<i>p</i> -value	0.021	0.004	0.003
Sample	Recorded	Omit zero spells	Omit Haiti/Togo
Full controls	Yes	Yes	Yes
Country and year effects	Yes	Yes	Yes
Observations	1,907	1,887	1,869
Countries	101	101	99
Within R^2	0.058	0.038	0.039

Notes: All columns include economic, institutional, and political controls and country and year effects. Column (1) uses OECD rates as recorded. Column (2) uses the main sample, excluding the twenty zero-rate records in Haiti and Togo described above. Column (3) excludes both countries entirely. Parentheses contain country-clustered standard errors. No replacement rates are imputed.

C.3 Treaty Exposure and Investment

The investment series is V15110 in Eurostat’s inward foreign affiliates statistics. It records gross investment in tangible goods, or annual additions to tangible assets, in millions of euros. These are expenditures by foreign affiliates rather than an FDI flow or a stock of assets. I organize the observations by host and ultimate controlling parent’s country. The treatment group consists of pairs in the 2020 intra-EU termination agreement with a treaty in force immediately before the judgment in 2018. Such treaties in the sample were all in force by 2010. Both countries must fall under the agreement, and I remove pairs with a previously terminated treaty instead of using them as controls.

The estimating equation for host h , parent country j , and year t is

$$\begin{aligned} \mathbb{E}[\text{Investment}_{hjt} \mid h, j, t] = & \exp\left(\alpha_{hj} + \lambda_{ht} + \lambda_{jt} + \delta_{hj}(t - 2017) \right. \\ & \left. + \sum_s \beta_s \text{Covered treaty}_{hj} \mathbf{1}\{t = s\}\right). \end{aligned} \quad (50)$$

I estimate (50) by Poisson pseudo-maximum likelihood. The term $\delta_{hj}(t - 2017)$ allows a separate linear time trend for each host-parent country pair and is included only in column (2).

The identifying comparison is between pairs with different treaty coverage, conditional on pair, host-year, and parent-year effects and, in column (2), country-pair linear trends. A causal interpretation requires the remaining investment shocks to be unrelated to coverage. Although hosts did not individually choose the date of the judgment, their past treaty choices or subsequent shocks specific to covered pairs may violate this condition. The exposure measure is also imperfect. An affiliate recorded under its ultimate parent’s country in Eurostat may seek arbitration through an intermediate holding company with a different treaty nationality.

Standard errors are clustered by host and parent country. The joint pre-event tests use host clustering because the two-way covariance matrix is not positive definite.

Reporting coverage falls sharply in 2018 and recovers in 2019 (Figure 4). Table 8 excludes 2018 and compares all available observations with pairs reporting in every remaining year, with and without country-pair linear trends.

The 2019 coefficients are negative in all four specifications. They are significant only in the complete-coverage samples, which also exhibit differential pre-event movements. These checks therefore support a descriptive interpretation of the investment decline.

TABLE 7: Annual investment coefficients around the Achmea judgment

	(1)	(2)
2010	−0.344 (0.271)	–
2011	−0.026 (0.284)	0.242 (0.314)
2012	−0.100 (0.181)	0.182 (0.219)
2013	−0.173 (0.257)	0.059 (0.235)
2014	−0.278* (0.138)	−0.102 (0.150)
2015	−0.172 (0.203)	−0.049 (0.127)
2016	−0.234*** (0.080)	−0.137 (0.111)
2018	−0.695*** (0.215)	−0.919*** (0.260)
2019	−0.253 (0.251)	−0.376 (0.346)
Country-pair linear trends	No	Yes
Observations	2,622	2,547

Notes: Full annual treatment coefficients for the two regressions in Table 2. All columns include pair, host-year, and parent-year effects. Reference years are 2017 in column (1) and 2010 and 2017 in column (2). Dashes denote terms not estimated. Parentheses contain standard errors clustered by host and parent country. ***, **, and * denote significance at 1, 5, and 10 percent.

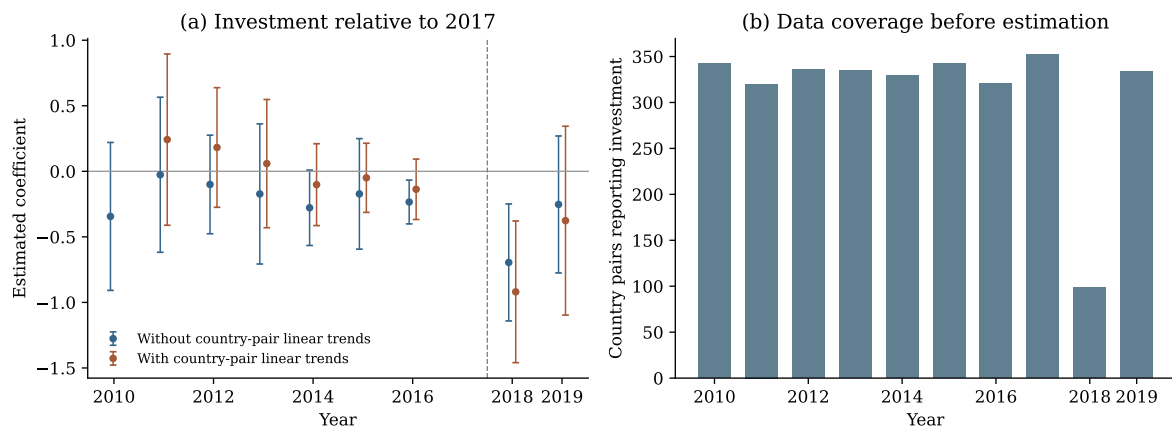


FIGURE 4: Investment around Achmea and the coverage of the data

Notes: Panel (a) reports annual coefficients and 95 percent confidence intervals for the specifications in Table 7. The vertical line marks the judgment. Panel (b) reports the number of country pairs with investment data in each year.

TABLE 8: Achmea estimates excluding the 2018 reporting gap

Country-pair linear trends	Available No	Complete No	Available Yes	Complete Yes
2019 treaty coefficient	-0.252	-0.523	-0.373	-0.662
Standard error	(0.250)	(0.185)	(0.348)	(0.237)
<i>p</i> -value	0.325	0.011	0.296	0.012
Joint pre-event <i>p</i> -value	0.018	< 0.001	0.546	< 0.001
Observations	2,527	1,674	2,478	1,674
Host countries	22	20	22	20

Notes: Annual event-study regressions estimated by Poisson pseudo-maximum likelihood over 2010–2019 excluding 2018, with country-pair, host-year, and parent-year effects. Complete samples contain pairs observed in all nine remaining years. The specifications follow Table 7. Parentheses contain standard errors clustered by host and parent country. Joint pre-event tests use host clustering.